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ANALYSIS OF HETEROGENEOUS CELL/TISSUE SAMPLES USING A MICROFLUIDIC DEVICE

Abstract

A microfluidic device for analyzing a sample comprising one or more sample objects containing biological cells, a method of analyzing a sample comprising one or more sample objects containing biological cells using such a microfluidic device and a system for analyzing a sample comprising one or more sample objects containing biological cells for use with such a microfluidic device. The one or more sample objects comprise one or more single cells, one or more cell aggregates and/or one or more pieces of continuous tissue material. The microfluidic device comprises an insertion volume configured to receive the sample. The microfluidic device further comprises a sorting unit having an inlet that is in fluid communication with the insertion volume, a first outlet and a second outlet. The sorting unit is configured to sort the sample objects by size, weight and/or stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, from the inlet towards the first outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, from the inlet towards the second outlet. The microfluidic device also comprises a measurement volume that is in fluid communication with the first outlet of the sorting unit. The microfluidic device further comprises a dissociation unit comprising means for dissociating sample objects into single cells and/or cell aggregates at least in part. An inlet of the dissociation unit is in fluid communication with the second outlet of the sorting unit and an outlet of the dissociation unit is in fluid communication with an inlet of the measurement volume.

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Background/Summary

FIELD

[0001] The present invention is in the field of biomedical research and clinical diagnostics. In particular, the invention relates to a microfluidic device for analyzing a sample comprising one or more sample objects containing biological cells, a method of analyzing a sample comprising one or more sample objects containing biological cells and a system for analyzing a sample comprising one or more sample objects containing biological cells.

BACKGROUND

[0002] Medical diagnoses are often based on a pathological analysis of tissue and cell morphology in samples obtained from a patient. A variety of sample types may be used for this, ranging from biopsies and aspirates from solid tissues to collection of fluids with single cells and/or small cell aggregates (e.g. effusions, urine) and blood samples. The morphology of cells in a sample may for example be captured by conventional microscopy. To analyze single cells from a tissue typically requires complex and time-consuming preparation procedures. These may involve the dissociation of larger tissue chunks to be further disassembled into single cells and/or cell aggregates, see e.g. D. Soteriou et al., medRxiv 2021.11.30.21267075 and US 2011/0003324 A1, completed by a final labelling step (e.g. staining) for optimal microscopic visualization. The cell labelling may be specifically adapted to diagnostic/scientific needs (e.g. standard stains like hematoxylin and eosin (H&E) and periodic acid-Schiff (PAS), fluorescence, in situ hybridization (ISH), immunohistochemistry (IHC)). This has so far prevented high-throughput real-time analyses of such samples in a clinical setting.

[0003] An imaging technique that holds great potential for studying the morphology of single cells and cells in cell aggregates is quantitative phase-contrast microscopy and in particular digital holographic microscopy (DHM). DHM uses the interference between an imaging beam and a reference beam to obtain phase and amplitude information of light transmitted by a sample. This information can then be used to reconstruct a quantitative phase shift image of the sample, see e.g. EP 1 524 491 A1 and EP 2 357 539 A1. In recent years, digital holographic microscopes have successfully been used in biomedical applications such as live cell imaging. Phase shift images of cells may be used to reliably identify cell types based on the analysis of morphological parameters and/or using machine learning classifiers. In combination with microfluidic systems, this allows for example high-throughput label-free blood sample analyses such as blood cell counts, see e.g. US 2019/0195774 A1, facilitating the detection of diseases like malaria, leukemia, and myeloproliferative neoplasms, see for example M. Ugele et al., Adv. Sci. 1800761 (2018), WO 2019/063548 A1, and M. Ugele et al., Proc. SPIE 11060, Optical Methods for Inspection, Characterization, and Imaging of Biomaterials IV, 110600V (2019). In this way, flow cytometry

can be used at the point-of-care for functional biomarker analysis based on cell types, cell morphology, cell-cell interactions, and kinetics of tissue dissociation. This solution cannot be obtained by fluorescence flow cytometry or related imaging flow cytometry methods due to the necessity of time-consuming labelling and thus the pre-defined sample volume with labeled analytes. With label-free DHM any sample volume with tissue, cells or cell aggregates can be visualized to achieve sufficient statistical power only limited by the amount of starting biopsy material. Even complex samples like solid tumors, composed of different cell types (i.e. epithelia, mesenchymal cells, and immune cells), and the tumor microenvironment are promising targets of the DHM method.

[0004] While digital holographic microscopes may be used to study cells in a liquid sample or in cell cultures including two-dimensional or three-dimensional tumor models, it is not suitable for studying larger sample objects such as pieces of continuous tissue material, e.g. from a biopsy sample. However, the latter constitute an important resource for the diagnosis and therapy of many diseases, e.g. pancreatic cancer. In the case of pancreatic lesions, for example, biopsies or fine needle aspirates are obtained by ultrasound-guided endoscopic sampling (e.g. endoscopic ultrasound-guided fine-needle biopsy (EUS-FNB) or endoscopic ultrasound-guided fine-needle aspiration (EUS-FNA)). Biopsies and fine-needle aspirates (FNA) will then be processed in histo- and cytopathology laboratories for conventional microscopic analyses to allow a morphology-based diagnosis. The post-sampling procedures are time consuming and costly and may require separate workflows. Moreover, for about 30% of patients—depending on body part or organ—surgical errors may occur as the lesion is not hit when taking the biopsy, see e.g. M. C. Omura, *American Journal of Roentgenology*, 197 (2), 457-461 (2011). In these cases, cellular material may be scarce, such that comprehensive tumor cell typing using immunohistochemistry cannot be applied, leading to inconclusive results not taking diagnostic errors into account.

SUMMARY

[0005] It is thus an object of the invention to provide means for analyzing heterogeneous cell/tissue samples, which may comprise single cells, cell aggregates as well as pieces of continuous tissue material, on the single cell level, in particular using a digital holographic microscope.

[0006] This object is met by a microfluidic device for analyzing a sample comprising one or more sample objects containing biological cells, a method of analyzing a sample comprising one or more sample objects containing biological cells and a system for analyzing a sample comprising one or more sample objects containing biological cells. Embodiments of the present invention are detailed in the dependent claims.

[0007] The microfluidic device according to the invention is configured for analyzing a sample comprising one or more sample objects containing biological cells, wherein the one or more sample objects comprise one or more single cells, one or more cell aggregates and/or one or more pieces of continuous tissue material. The microfluidic device comprises an insertion volume configured to receive the sample. The microfluidic device further comprises a sorting unit having an inlet that is in fluid communication with the insertion volume, a first outlet and a second outlet. The sorting unit is configured to sort the sample objects by size, weight and/or stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, from the inlet towards the first outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, from the inlet towards the second outlet. The microfluidic device comprises a measurement volume that is in fluid communication with the first outlet of the sorting unit. The microfluidic device further comprises a dissociation unit comprising means for dissociating sample objects into single cells and/or cell aggregates at least in part. An inlet of the dissociation unit is in fluid communication with the second outlet of the sorting unit and an outlet of the dissociation unit is in fluid communication with an inlet of the measurement volume.

[0008] As used herein, continuous tissue material may refer to a tightly-joined structure of a large number of cells, which may e.g. comprise the cells and an extracellular matrix linking the cells to

each other. A piece of continuous tissue material, which may also be referred to as a piece of tissue in the following, may for example be a biopsy sample or a small chunk of tissue from an aspiration sample. A piece of continuous tissue material may for example have a physical dimension between 0.5 mm and 2 cm in diameter and/or length and/or a volume between $10.\text{sup.}-3\text{ mm}^3$ and $8\text{ mm}.\text{sup.}3$. A cell aggregate, on the other hand, may refer to a plurality of cells that are loosely bound or adhere to each other, but without forming a continuous tissue material. A cell aggregate may for example have a physical dimension between $1\text{ }\mu\text{m}$ to $500\text{ }\mu\text{m}$ in every direction and/or a volume between $3\text{ }\mu\text{m}.\text{sup.}3$ and $108\text{ }\mu\text{m}.\text{sup.}3$. A cell aggregate may for example be a cluster of cells comprising between 2 and 100 cells, in some examples also more than 100 cells.

[0009] The microfluidic device may for example be or comprise a microfluidic chip, wherein the microfluidic chip may e.g. comprise a substrate in which the insertion volume, the sorting unit, the measurement volume, and/or the dissociation unit are formed. In some embodiments, the microfluidic device may be modular and may for example comprise a plurality of modules, some or all of which may be replaceable, e.g. to prevent cross-contamination of samples. For example, the insertion volume, the sorting unit, the measurement volume and/or the dissociation unit may each be contained in such a module, e.g. such that some (e.g. the reservoirs) or all of the aforementioned components can be replaced individually. The microfluidic device may further comprise a plurality of channels connecting the aforementioned components, for example such that the insertion volume is arranged upstream of the sorting unit and the dissociation unit and the measurement volume is arranged downstream of the sorting unit and the dissociation unit, respectively. The microfluidic device may also comprise one or more microfluidic components such as one or more pumps and/or one or more valves for generating and controlling fluid flows within the microfluidic device. The microfluidic chip may further comprise one or more reservoirs for storing fluids such as a carrier fluid in which the sample is to be dissolved or suspended, e.g. by supplying the carrier fluid to the insertion volume.

[0010] The microfluidic device may be configured to transfer or transport sample objects from a sample arranged in the insertion volume to the sorting unit, e.g. by generating a flow of a carrier fluid. The sample objects may be sorted by size, weight and/or stiffness in the sorting unit with smaller, lighter and/or less stiff sample objects being directly transferred to the measurement volume via the first outlet of the sorting unit, whereas larger and/or heavier sample objects may be transferred to the dissociation unit via the second outlet of the sorting unit for dissociating the sample objects prior to being transferred to the measurement volume. In the measurement volume, one or more measurements may be performed on the sample objects, e.g. as detailed below for the method according to the invention.

[0011] Thereby, the present invention allows for analyzing a heterogeneous sample comprising sample objects of various sizes, e.g. pieces of tissue as well as single cells and cell aggregates, and/or different types of samples such as biopsy samples and body fluid samples using the same device and similar workflows. For example, when placing a biopsy sample in the insertion volume, only blood cells (e.g. from the extraction of the biopsy sample) and/or mobile cells such as mesenchymal cells may directly pass to the measurement volume, whereas the biopsy sample itself and/or larger cell aggregates may be transferred to the dissociation unit for dissociation. The biopsy sample may be dissociated sequentially, e.g. by enzymatic digestion. This may for example allow for implementing assays to study tissue composition, e.g. by slowly dissolving the biopsy sample from the outside to the inside.

[0012] The sorting unit is configured to sort the sample objects by size, weight and/or stiffness, e.g. using a mechanical filter structure and/or by hydrodynamic and/or by inertia-based sorting as detailed below. As used herein, the size of a sample object may for example refer to a physical dimension of the respective sample object such as a length, a width, a diameter or an effective diameter or to a volume of the sample object. The sorting unit may for example be configured to sort the sample objects such that the sample objects directed towards the second outlet are larger

(e.g. have a larger average size), heavier (e.g. have a larger average weight) and/or stiffer (e.g. having a higher average stiffness and/or a lower average elasticity) than the sample objects directed towards the first outlet. In some embodiments, the sorting unit may further be configured to separate the sample objects from a fluid or medium (or a part thereof) that the sample objects are contained or suspended in (e.g. a carrier fluid and/or a dissolving solution), for example to exchange said fluid or medium by a different fluid or medium (e.g. a measurement fluid). For this, the sorting unit may for example be configured to generate an accelerating force on the sample objects and the fluid/medium, e.g. similar to a centrifuge, and/or apply principles, such as lateral displacement, hydrodynamics and/or cross flow approaches. Preferably, the sorting unit is configured to sort the sample objects at least by size and/or weight (and, optionally, by stiffness in addition). In some embodiments, the sorting unit may comprise multiple sorting stages, e.g. a first sorting stage for sorting the sample objects by size and/or weight (e.g. into the first and second outlets) and a second sorting stage for sorting the sample objects by stiffness (e.g. at or downstream of the first outlet and/or the second outlet, for example to filter out dead and/or non-viable cells).

[0013] The sorting unit may for example be configured to direct sample objects having a size and/or a weight within a respective first range towards the first outlet and to direct sample objects having a size and/or a weight within a respective second range towards the second outlet, wherein an upper bound of the first range is smaller than or equal to a lower bound of the second range. In one example, the sorting unit is configured to direct sample objects having a size and/or a weight smaller/lighter than a respective sorting threshold towards the first outlet and to direct sample objects having a size and/or a weight larger/heavier than the respective sorting threshold towards the second outlet. The sorting threshold may for example be between 5 μm and 1000 μm , in some examples between 5 μm and 200 μm , in some examples between 10 μm and 100 μm , in one example between 30 μm and 70 μm . The sorting unit may for example be configured to separate single cells and small cell aggregates (e.g. cell aggregates with a size up to the sorting threshold) from larger cell aggregates and pieces of tissue. The sorting threshold may be chosen depending on the type of sample, physical dimensions of the measurement volume and/or a measurement technique used for performing measurements in the measurement volume. For a sample containing larger sample objects such as pieces of tissue (e.g. a biopsy sample), a larger sorting threshold may be used (e.g. 100 μm or 1000 μm), whereas for a sample containing smaller sample objects such as a single cells and/or small cell aggregates (e.g. a body fluid sample), a smaller sorting threshold may be used (for example 50 μm or 100 μm).

[0014] The stiffness of a sample object may for example characterize or quantify an extent to which the respective sample object resists deformation when applying a force to the sample object. The stiffness may for example be quantified by a ratio of the applied force to a displacement (e.g. a change in a physical dimension such as the length of the sample object) or by a related quantity such as an elastic modulus (e.g. Young's modulus). The sorting unit may for example be configured to sort the sample objects by stiffness based on a threshold and/or first and second ranges for the respective quantity, e.g. as detailed above for the sorting by size and/or weight. In one example, the sorting unit is configured to sort the sample objects by stiffness by directing sample objects having a Young's modulus smaller than a (stiffness) sorting threshold towards the first outlet and sample objects having a Young's modulus larger than said sorting threshold towards the second outlet, wherein said sorting threshold may for example be between 50 Pa and 30 kPa, in some examples between 210 Pa and 23 kPa, in some examples between 400 Pa and 10 kPa. In one example, said sorting threshold is between 200 Pa and 400 Pa, e.g. 300 Pa. Sorting of sample objects by stiffness may for example be implemented as described in G. Wang et al., PLOS ONE 8(10), e75901 (2013).

[0015] The dissociation unit may comprise one or more dissociation volumes, in which the sample objects are to be dissociated, e.g. dissolved chemically (for this document including chemical and/or enzymatic means) and/or dissociated mechanically (e.g. as a result of a force acting on the respective sample object). The dissociation volumes may be arranged between the inlet and the

outlet of the dissociation unit. The dissociation volumes may be arranged in series, e.g. such that a first dissociation volume is arranged upstream of a second dissociation volume such that sample objects flowing through the dissociation unit first pass through the first dissociation volume and then through the second dissociation volume. Additionally or alternatively, the dissociation volumes may be arranged in parallel, e.g. such that sample objects flowing through the dissociation unit pass through either the first dissociation volume or the second dissociation volume.

[0016] The dissociation unit comprises means for dissociating sample objects into single cells and/or cell aggregates at least in part. The means for dissociating sample objects may be configured to remove single cells and/or cell aggregates from a sample object, e.g. mechanically, chemically, thermally and/or electrically, and/or to separate or break apart a sample object into two or more smaller sample objects, e.g. into smaller pieces of tissue and/or smaller cell aggregates. As used herein, mechanical dissociation may refer to any dissociation as a result of a force acting on the respective sample object, for example by cutting, grinding and/or sonication.

[0017] The means for dissociating sample objects may comprise a reservoir configured to supply a dissolving solution to a dissociation volume. The dissolving solution may comprise one or more chemical and/or biochemical substances for chemically dissolving sample objects. The dissolving solution may in particular be or comprise an enzymatic solution, wherein the enzymatic solution may comprise one or more enzymes for dissolving sample objects by enzymatic digestion. The one or more chemical and/or biochemical substances, e.g. the one or more enzymes, may be configured to break down macromolecules and may for example comprise one or more proteases for breaking down proteins and/or one or more nucleases for breaking down nucleic acids. The one or more chemical and/or biochemical substances/enzymes may for example be selected from the group consisting of trypsin, collagenase (e.g. collagenase type II), DNase, accutase, dispase, papain, liberase, pronase, protease, proteinase (K), thrombin and ethylenediaminetetraacetic acid (EDTA). The means for dissociating sample objects may further comprise a pump and/or a valve for supplying the dissolving/enzymatic solution from the reservoir to the dissociation volume.

[0018] Additionally or alternatively, the means for dissociating sample objects may comprise an acoustic emitter configured to apply acoustic waves (e.g. sound waves) to sample objects in a dissociation volume, in particular an ultrasound emitter configured to apply ultrasound to sample objects in a dissociation volume. The acoustic/ultrasound emitter may for example be or may comprise a passive emitter configured to transfer acoustic waves/ultrasound generated by an external acoustic/ultrasound source such as a sonotrode to the dissociation volume. A shape, a size and/or a material of the acoustic/ultrasound emitter may be adapted to increase the transmission efficiency into the dissociation volume. For example, a resonance frequency of the ultrasound emitter may be adapted to a frequency of the ultrasound to be applied to the microfluidic device and may for example be between 20 kHz and 40 kHz. Additionally or alternatively, the acoustic/ultrasound emitter may be or may comprise an active emitter configured to generate acoustic waves/ultrasound, e.g. an acoustic/ultrasonic transmitter such as a piezo element, a capacitive element and/or an inductive element that is configured to convert an electrical signal applied to the acoustic/ultrasound emitter to acoustic waves/ultrasound.

[0019] Additionally or alternatively, the means for dissociating sample objects may comprise one or more static obstacles arranged in a dissociation volume configured to mechanically dissociate sample objects colliding with the one or more obstacles. Some or all of the static obstacles may for example comprise a sharp element such as a sharp corner of the respective static obstacle or a blade that is configured to cut or slice through a sample object. In some embodiments, the static obstacles may be arranged in a one-dimensional or two-dimensional grid, for example in a grid that extends perpendicular to a direction of flow in the dissociation volume, e.g. such that the static obstacles are displaced from each other in a direction perpendicular to the direction of flow.

[0020] Additionally or alternatively, the means for dissociating sample objects may comprise one or more curved and/or constricted portions of a dissociation volume configured to dissociate

sample objects by generating a shear force on sample objects flowing through the dissociation volume. A width, a height and/or a cross-sectional area of the dissociation volume perpendicular to a direction of flow through the dissociation volume may for example vary along the direction of flow, e.g. such that the dissociation volume comprises a plurality of constricted portions separating non-constricted portions having a larger width, height and/or cross-sectional area than the adjacent constricted portions. A width, a height and/or a cross-sectional area of a constricted portion may for example be less than 75%, in some examples less than 50%, in one example less than 25% of the respective quantity of an adjacent non-constricted portion upstream of the constricted portion. A width and/or a height of a constricted portion may for example be between 100 μm and 2 mm. In one example, the means for dissociating sample objects comprise a plurality of constricted portion arranged in series, wherein a width, a height and/or a cross-sectional area of the constricted portions may decrease successively along the direction of flow in some examples (e.g. such that larger constricted portions are arranged upstream of smaller constricted portions). Additionally or alternatively, the dissociation volume may for example comprise one or more meandering channels, each of which may comprise a plurality of curved portions. A radius of curvature of a sidewall of the dissociation volume in the curved and/or constricted portions and/or the physical dimensions and/or arrangement of the constricted portions may be chosen such that the shear force on sample objects flowing through the dissociation volume is sufficiently large to break up or tear apart the sample objects at least in part.

[0021] Additionally or alternatively, the means for dissociating sample objects may comprise a stirrer, particularly a magnetic stirrer, configured to move a stir bar, particularly a magnetic stir bar, in a dissociation volume for dissociating sample objects. The magnetic stirrer may for example comprise one or more inductive elements such as coils that are configured to generate a time-varying magnetic field, for example a rotating magnetic field for rotating the magnetic stir bar. The inductive elements may for example be arranged on or in a sidewall of the dissociation volume. The means for dissociating sample objects may also comprise the stir bar, which may be arranged in the dissociation volume. The stir bar may comprise a sharp element such as a blade or a sharp corner that is configured to cut or slice through a sample object.

[0022] Additionally or alternatively, the means for dissociating sample objects may comprise one or more moveable elements configured to cut and/or grind sample objects in a dissociation volume. The moveable elements may for example comprise one or more blades, which may e.g. be arranged in or on a sidewall of the dissociation volume and may be configured to be rotated in the dissociation volume and/or moved back and forth in the dissociation volume, e.g. into and out of the dissociation volume, to cut or slice through a sample object. Additionally or alternatively, the moveable elements may comprise one or more grinders, each of which may e.g. comprise one or more rotatable disks which may be equipped with protrusions such as teeth to grind a sample object.

[0023] In some embodiments, the dissociation unit comprises a plurality of dissociation volumes, each of which is associated (e.g. equipped) with a different means for dissociating sample objects. For example, the dissociation unit may comprise a first dissociation volume associated with a first means for dissociating sample objects and a second dissociation volume associated with a second means for dissociating sample objects, wherein each of the first and second means for dissociating sample objects may be embodied as detailed above. The first means may for example be a reservoir configured to supply a dissolving solution to the first dissociation volume and the second means may for example be an ultrasound emitter configured to apply ultrasound to the second dissociation volume. In some examples, the dissociation unit may comprise at least three dissociation volumes with three different means for dissociating sample objects (e.g. a reservoir for a dissolving solution, an ultrasound emitter and one or more blades), in one example four dissociation volumes with four different means for dissociating sample objects (e.g. a reservoir for a dissolving solution, an ultrasound emitter, a first set of blades and a second set of blades). The dissociation volumes may

be arranged in series and/or in parallel as detailed above. In some embodiments, some or all of the dissociation volumes may be associated with more than one means for dissociating sample objects. For example, one or more of the dissociation volumes may be associated with a mechanical means and a chemical means (e.g. a reservoir for a dissolving solution) for dissociating sample objects. In one embodiment, the dissociation unit comprises a common reservoir for a dissolving solution, wherein the common reservoir is configured to supply the dissolving solution to two or more, in one example to all of the dissociation volumes, e.g. via an inlet upstream of the dissociation volumes.

[0024] In a preferred embodiment, the dissociation unit further comprises a plurality of pumps and/or valves for selectively directing sample objects into one of the dissociation volumes, e.g. one pump and/or one valve for each of the dissociation volumes. Additionally or alternatively, the sorting unit may be configured to selectively direct sample objects into one of the dissociation volumes based on a size, a weight and/or a stiffness of the respective sample object, e.g. as detailed below.

[0025] The sorting unit may comprise one or more of an inertia-based cell filter, an acoustic cell filter, a hydrodynamic cell filter, a dead-end cell filter, a cell-stiffness sorter and a cross-flow cell filter. Additionally or alternatively, the sorting unit may also comprise a viscoelastic filter, a density sorter, a field-flow fractionation filter, a dielectrophoretic filter, a deterministic lateral displacement filter, and/or a weir filter.

[0026] An inertia-based cell filter may for example be configured to deflect a flow (i.e. induce a change in a direction of flow) of a carrier fluid containing the sample objects for sorting the sample objects. The inertia-based cell filter may e.g. comprise a channel with one or more curved portions, e.g. a spiral-shaped channel. The size and/or the weight of the sample objects may determine the extent to which the sample objects can follow the change in the flow direction. For example, lighter and/or smaller sample objects may be able to follow the deflected flow, whereas heavier and/or larger sample objects may not or only partially be able to follow the deflected flow, e.g. as a result of a centrifugal force acting on the sample objects. The first and second outlets of the sorting unit may be arranged such that smaller and/or lighter sample objects end up in the first outlet, whereas larger and/or heavier sample objects end up in the second outlet. A sorting range or threshold of the inertia filter may be controlled by adjusting a flow velocity in the inertia-based cell filter. Similarly, a hydrodynamic cell filter may be configured to sort the sample objects by generating size- and/or weight-dependent hydrodynamic forces on the sample objects.

[0027] An acoustic cell filter may for example be configured to sort the sample objects by generating a standing acoustic wave, e.g. a standing ultrasound wave. The acoustic wave may exert size-dependent forces on the sample objects, which may be used to deflect the sample objects into either the first or second outlet of the sorting unit depending on their size.

[0028] A dead-end cell filter and a cross-flow cell filter may comprise a mechanical filter structure such as a grid, a mesh, a lattice structure or a plate comprising a plurality of holes or perforations. The mechanical filter structure may be configured to prevent sample objects with a size larger than a sorting threshold (e.g. the size of the holes) from passing through the mechanical filter structure. In a dead-end cell filter, a carrier fluid containing the sample objects may be forced through the mechanical filter structure (thereby sorting out sample objects larger than the sorting threshold, which cannot pass through the filter structure), whereas in a cross-flow cell filter the carrier fluid may flow parallel to a surface of the mechanical filter structure (thereby sorting out sample objects smaller than the sorting threshold, which can pass through the filter structure and thus leave the carrier fluid flow).

[0029] The second outlet of the sorting unit may comprise a plurality of sub-outlets (e.g. between two and ten sub-outlets), for example to further sort the sample objects directed towards the second outlet similar to the sorting into the first and second outlets described above. For example, the sorting unit may be configured to sort the sample objects directed towards the second outlet by

size, weight and/or stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, towards a first sub-outlet of the second outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, towards a second sub-outlet of the second outlet, e.g. similar as described above. Each of the sub-outlets of the second outlet may for example be associated with a respective sorting threshold (e.g. a size and/or weight threshold) and/or a respective sorting range (e.g. a size and/or weight range).

[0030] In some embodiments, the sorting unit may comprise a waste outlet, e.g. to sort out dead and/or non-viable cells. The sorting unit may for example be configured to sort the sample objects by stiffness by directing sample objects having a higher stiffness from the inlet towards the waste outlet and sample objects having a lower stiffness from the inlet towards the first outlet and/or towards the second outlet (e.g. as a second sorting stage after sorting by size and/or weight). The waste outlet may for example be in fluid communication with a waste reservoir or container.

[0031] In some embodiments, the sorting unit may be configured to selectively direct sample objects into one of the dissociation volumes based on their size, weight and/or stiffness. For example, each of the sub-outlets of the second outlet may be associated with a respective means for dissociating sample objects and/or a respective dissociation volume, e.g. for dissociating sample objects by different means based on their size, weight and/or stiffness. For example, the first sub-outlet may be in fluid communication with a first dissociation volume, but not with a second dissociation volume, whereas the second sub-outlet may be in fluid communication with the second dissociation volume, but not with the first dissociation volume.

[0032] Similarly, the first outlet of the sorting unit may also comprise a plurality of sub-outlets to further sort the sample objects directed towards the first outlet by size, weight and/or stiffness. Each of the sub-outlets of the first outlet may for example be in fluid communication with a different measurement volume, e.g. a first sub-outlet with a first measurement channel for smaller sample objects and a second sub-outlet with a second measurement channel for larger sample objects as described below.

[0033] In a preferred embodiment, the microfluidic device is configured to sort sample objects dissociated in the dissociation unit again prior to transfer to the measurement volume, e.g. to prevent larger sample objects from entering the measurement volume and/or to perform additional dissociation steps for dissociating larger sample objects further. For this, the outlet of the dissociation unit may for example be in fluid communication with the measurement volume via the sorting unit. In other words, the dissociated sample objects may have to pass through the sorting unit again with only sample objects directed towards the first outlet thereof being transferred to the measurement chamber. For example, a channel from the outlet of the dissociation volume may intersect with a channel connecting the insertion volume to the sorting unit upstream of the sorting unit.

[0034] Additionally or alternatively, the outlet of the dissociation unit may further comprise a sorter (e.g. an additional sorting unit), which may be similar or identical to the sorting unit described above. The sorter may have a first outlet that is in fluid communication with the measurement volume. The sorter may further have a second outlet that is in fluid communication with the inlet of the dissociation unit. The sorter may be configured to sort the sample objects by size, weight and/or stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, towards the first outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, towards the second outlet, e.g. as described above. In one example, the second outlet of the sorter comprises a plurality of sub-outlets, e.g. as described above for the sorting unit. In some embodiments, the second outlet of the sorter may not be in fluid communication with the measurement volume, but may for example be connected to a waste outlet or a waste reservoir of the microfluidic device, e.g. for discarding sample objects that have not been dissociated sufficiently.

[0035] In some embodiments, the microfluidic device further comprises a filter arranged between

the outlet of the dissociation unit and the inlet of the measurement volume. Additionally or alternatively, said filter (or an additional filter similar to said filter) may be arranged between the first outlet of the sorting unit and the inlet of the measurement volume. The filter may for example be arranged at the inlet of the measurement volume or along a channel connecting the outlet of the dissociation unit to the inlet of the measurement volume, preferably along a common channel connecting the inlet of the measurement volume to the first outlet of the sorting unit and to the outlet of the dissociation unit. The filter may be configured to prevent sample objects having a size larger than a filtering threshold from entering the measurement volume. The filter may for example be or comprise a dead-end cell filter or a cross-flow cell filter as described above. Additionally or alternatively, the filter may be configured to filter out dead or non-viable cells. The filter may for example be or comprise a cell-stiffness sorter that is configured to filter or sort sample objects by stiffness, e.g. as described above for the sorting unit. This may for example allow to filter out dead or non-viable cells, which may exhibit a higher stiffness than living or viable cells, for example as described in M. Islam et al., Scientific Reports 7, 1997 (2017). In some embodiments, the filter may also be configured to exchange a fluid or medium (or a part thereof) that the sample objects are contained or suspended in, for example to exchange a carrier fluid and/or a dissolving solution by a different fluid or medium such as a measurement fluid. In one example, said filter or parts thereof may be comprised in the sorting unit.

[0036] The microfluidic device may comprise a sample collection unit for inserting a sample into the insertion volume. The sample collection unit may comprise one or more means for inserting a sample into the insertion volume, each of which may be configured to interface with a sample collection device. The sample collection unit may for example comprise a connector configured to couple a syringe and/or a tube containing a sample (e.g. a sample fluid) comprising the one or more sample objects to the insertion volume. Additionally or alternatively, the sample collection unit may comprise an opening configured to receive an aspiration needle, e.g. a fine-needle aspiration (FNA) needle, and/or a biopsy needle, e.g. a fine-needle biopsy (FNB) needle, wherein the opening is in fluid communication with the insertion volume.

[0037] Additionally or alternatively, the sample collection unit may comprise a transfer pump configured to transfer a sample from a collection tube (e.g. a standard centrifuge tube) to the insertion volume. The transfer pump may for example be configured to suck the sample out of the collection tube and to release the sample into the insertion volume. The transfer pump may also be configured to flush the collection tube, e.g. to detach a sample from a sample collection device such as a biopsy needle, a brush or a spatula. The collection tube may be part of the microfluidic device or may be a separate element. The collection tube may be removable or may be permanently integrated into the microfluidic device. In some examples, the transfer pump may be arranged in a moveable head. The moveable head may be configured to be inserted into the collection tube, e.g. for sucking a sample out of the collection tube using the transfer pump. The moveable head may further be configured to move the sample to the insertion volume, where the sample may be released, e.g. flushed out of the moveable head by the transfer pump.

[0038] The microfluidic device may further comprise a wiper for wiping off a sample from a sample collection device, in particular for wiping off a biopsy sample from a biopsy needle. A wiper may for example be arranged on a side wall of the insertion volume and/or on a side wall of the collection tube. The wiper may e.g. be a protrusion or a step that the sample can be brought in contact with for detaching the sample from the sample collection device.

[0039] The microfluidic device may further comprise a flow pump configured to generate a flow through the insertion volume to transport the sample objects from the insertion volume to the sorting unit. The flow pump may also be configured to flush and/or suck out the insertion volume, e.g. to detach a sample from a sample collection device such as a brush or a spatula and/or from the wiper. The flow pump may in particular be configured to detach a biopsy sample from a biopsy needle and/or from the wiper by flushing and/or sucking out the insertion volume. In some

embodiments, the flow pump may also be the transfer pump.

[0040] In some embodiments, the measurement volume comprises a first measurement channel and a second measurement channel connected in parallel to the first measurement channel. The second measurement channel may have a larger cross-sectional area than the first measurement channel, e.g. for analyzing sample objects with a larger size. The cross-sectional area of the second measurement channel may for example be between 1.5 times and 5 times as large as the cross-sectional area of the first measurement channel. The microfluidic device may comprise a pump and/or a valve for selectively directing sample objects into either the first measurement channel or the second measurement channel. Additionally or alternatively, the sorting unit may be configured to selectively direct sample objects into either the first measurement channel (e.g. via a first sub-outlet of the first outlet of the sorting unit) or the second measurement channel (e.g. via a second sub-outlet of the first outlet of the sorting unit) based on a size, a weight and/or a stiffness of the respective sample object. The sorting unit may for example be configured such that sample objects having a smaller size, a smaller weight and/or a lower stiffness are directed into the first measurement channel and sample objects having a larger size, a larger weight and/or a higher are directed into the second measurement channel.

[0041] The microfluidic device may also comprise a measurement fluid inlet arranged upstream of the inlet of the measurement volume for supplying a measurement fluid. The measurement fluid inlet may in particular be arranged between the outlet of the dissociation unit and the inlet of the measurement unit. This may for example allow for replacing a carrier fluid used for transporting the sample objects through the sorting unit and/or through the dissociation unit, which may e.g. contain enzymes added in the dissociation unit, by a different fluid that measurements in the measurement volume are to be performed in, for example a viscoelastic fluid, and/or to add substances such as a viscoelastic substance to the carrier fluid.

[0042] In a preferred embodiment, the measurement volume comprises an optical detection window for taking microscopic images, in particular phase shift images of sample objects in the measurement volume. The detection window may have optical properties that allow for microscopic imaging and in particular phase shift imaging therethrough. The detection window may for example have a transmitted wavefront error of no more than $\lambda/2$, in some examples of no more than $\lambda/4$, in one example of no more than $\lambda/8$, wherein the transmitted wavefront error may e.g. be measured according to the ISO 10110 standard. In some embodiments, the detection window may have an anti-reflective coating.

[0043] The present invention further provides a method of analyzing a sample comprising one or more sample objects containing biological cells using a microfluidic device according to any one of the embodiments described therein. The one or more sample objects comprise one or more single cells, one or more cell aggregates and/or one or more pieces of continuous tissue material. The method comprises generating a flow of a carrier fluid through the insertion volume to transport sample objects from the insertion volume to the sorting unit. The sample objects are sorted with the sorting unit into a first batch of sample objects and a first residual sample. The first residual sample comprises sample objects having a larger size, a larger weight and/or a higher stiffness than the sample objects in the first batch. A first measurement is performed on the sample objects of the first batch in the measurement volume. Sample objects in the first residual sample are dissociated with the dissociation unit into single cells and/or cell aggregates at least in part to obtain a second batch of sample objects. A second measurement is performed on the dissociated sample objects in the second batch in the measurement volume.

[0044] The method may also comprise providing the sample in the insertion volume, e.g. via a connector or an opening as described above or by transferring the sample from a collection tube as described above. This may comprise wiping the sample off a sample collection device using a wiper and/or flushing and/or sucking out the insertion volume and/or the collection tube to detach the sample from a sample collection device. In other examples, the sample may be provided in the

insertion volume prior to executing the method. A carrier fluid such as phosphate-buffered saline may be provided in the insertion volume. A flow of the carrier fluid may be generated from the insertion volume through the sorting unit to the measurement volume and to the dissociation unit to transport some or all of the sample objects in the insertion volume to the sorting unit and further on to either the measurement volume (first batch) or the dissociation unit (first residual sample).

[0045] The sample objects may be sorted as described above for the microfluidic device according to the invention with the sample object in the first batch being directed towards the first outlet of the sorting unit (and then to the measurement volume) and the sample objects in the first residual sample being directed towards the second outlet of the sorting unit (and then to the dissociation unit). The sample objects may for example be sorted such that the first batch contains sample objects having a size and/or a weight within a respective first range and/or below a respective sorting threshold, whereas the first residual sample contains sample objects having a size and/or a weight within a respective second range and/or above the respective sorting threshold with an upper bound of the first range being smaller than or equal to a lower bound of the second range. The first batch may for example contain single cells and cell aggregates, but no pieces of tissue or may contain single cells and small cell aggregates, but no large cell aggregates and no pieces of tissue. Additionally or alternatively, the sample objects may also be sorted by stiffness, e.g. as described above.

[0046] The sample objects may be dissociated into smaller objects as described above using one or more means for dissociating sample objects, e.g. by chemical dissociation and in particular enzymatic digestion. For example, a dissolving solution such as an enzymatic solution comprising one or more enzymes for dissolving sample objects by enzymatic digestion may be supplied to a dissociation volume that the sample objects are arranged in or flowing through. The sample objects may be exposed to the dissolving solution for a predetermined amount of time, e.g. between 1 min and 60 min, in one example between 5 min and 30 min. Additionally or alternatively, the sample objects may be dissociated mechanically, for example by applying acoustic waves (e.g. ultrasound), by generating a shear force and/or by mechanically dissociating the sample objects by static obstacles and/or moveable elements such as a stir bar or a grinder. In some embodiments, the sample objects may be dissociated sequentially using different means for dissociating sample objects, for example by first dissociating the sample objects by a first means (e.g. cutting and/or grinding sample objects mechanically) and then further dissociating the sample objects by a second means (e.g. dissolving the sample objects by enzymatic digestion).

[0047] The second batch of sample objects may be obtained by sorting the dissociated sample objects into the second batch of sample objects and a second residual sample, e.g. similar to the sorting of the sample objects into the first batch and the first residual sample with the sorting unit described above. The second residual sample may comprise sample objects having a larger size, a larger weight and/or a higher stiffness than the sample objects in the second batch. The dissociated sample objects may for example be sorted using the sorting unit or using a sorter at an outlet of the dissociation unit. The second batch may then be transferred to the measurement volume for performing the second measurement.

[0048] The second residual sample may either be discarded or may be dissociated further to obtain one or more additional batches of sample objects, e.g. by performing one or more additional dissociation and/or sorting steps. An additional measurement may be performed on dissociated sample objects in each of the additional batches in the measurement volume. The one or more additional batches of sample objects may for example be obtained by dissociating sample objects in a residual sample separated from a previous batch (e.g. the second residual sample separated from the second batch) into single cells and/or cell aggregates at least in part using the dissociation unit and sorting the dissociated sample objects into the respective additional batch of sample objects (e.g. the third batch) and a new residual sample (e.g. the third residual sample). The new residual sample may comprise sample objects having a larger size, a larger weight and/or a higher stiffness

than the sample objects in the respective additional batch, e.g. as described above. The method may for example comprise obtaining between 1 and 20 batches, in one example between 3 and 10 batches of sample objects from the sample. In some embodiments, different means for dissociating sample objects may be used for different dissociation steps. For example, mechanical means may be used in a first dissociation step while chemical dissociation (e.g. enzymatic digestion), ultrasound and/or shear force may be used in one or more later dissociation steps.

[0049] In this way, the sample objects may be dissociated sequentially and separated into a plurality of batches. Each of the batches may for example only contain objects within a certain size range, e.g. only objects smaller than a sorting threshold. This may allow for performing measurements on the objects in each of the batches on the single cell level and may further prevent larger objects from clogging the measurement volume. Each of the batches may be associated with certain types of sample objects and/or certain parts of the sample objects and may thus provide unique information on the original sample and its composition. For example, the first batch may predominantly contain single cells and/or small cell aggregates that were already contained in the original sample and the second batch may predominantly contain single cells and/or small cell aggregates originating from larger cell aggregates and/or from a surface layer of pieces of tissue from the original sample. Subsequent batches may predominantly contain single cells and/or small cell aggregates originating from deeper and deeper layers of pieces of tissue from the original sample, thereby allowing for a spatially resolved analysis of the pieces of tissue.

[0050] In some embodiments, the method may comprise filtering some or all of the batches of sample objects before the respective batch enters the measurement volume, e.g. to prevent sample objects having a size larger than a filtering threshold from entering the measurement volume and/or to remove dead and/or non-viable cells. The batches may for example be filtered using a filter arranged upstream of the measurement volume as detailed above for the microfluidic device according to the invention.

[0051] The method may comprise determining a sample type of the sample, wherein the sample type characterizes one or more of a procedure used for obtaining the sample from a patient (e.g. biopsy, aspiration, resection, puncture, lavage or smear), a body part that the sample was obtained from (e.g. an organ, a type of tissue or a type of body fluid), a size of the sample objects in the sample (e.g. an average size and/or a maximum size) and a weight of the sample objects in the sample. The sample type may for example be determined based on a user input (the user may e.g. enter or specify the sample type using an input device such as a touchscreen or touchpad) and/or based on a label such as a barcode associated with the sample and/or a sample collection device. The method may further comprise determining a number of dissociation steps that are to be performed based on the sample type and performing the dissociation steps and sorting the dissociated sample objects by size, weight and/or stiffness to obtain a plurality of batches of sample objects. For example, the larger the size of the sample objects in the sample is the more dissociation steps may be performed.

[0052] The sample may for example comprises one or more of a biopsy sample (e.g. a fine-needle biopsy sample), an aspiration sample (e.g. a fine-needle aspiration sample), a body fluid sample (wherein the body fluid may e.g. be blood, lymph, an effusion (e.g. a pleural, peritoneal or pericardial effusion), cerebrospinal fluid, urine or combinations thereof), a smear sample (e.g. an oral smear, a cervical smear and/or a bone smear) and/or a resection sample (e.g. tissue pieces from an excision and/or resection such as sonicated brain tissue of resection rim).

[0053] A smaller number or no dissociation steps may be performed if the sample type indicates that the sample is a body fluid sample and/or if the sample type indicates that the sample is a smear sample, while a larger number of dissociation steps may be performed if the sample type indicates that the sample is a biopsy sample, if the sample type indicates that the sample is an aspiration sample and/or if the sample type indicates that the sample is a resection sample. In one example, no dissociation step or a single dissociation step is performed if the sample type indicates that the

sample is a body fluid sample and/or if the sample type indicates that the sample is a smear sample and multiple dissociation steps (e.g. between two and ten dissociation steps) are performed if the sample type indicates that the sample is a biopsy sample and/or if the sample type indicates that the sample is an aspiration sample. Similarly, a larger number of dissociation steps may be performed if the sample type indicates that the sample is a biopsy sample than if the sample type indicates that the sample is an aspiration sample.

[0054] In some embodiments, the dissociation unit of the microfluidic device comprises a plurality of means for dissociating the sample objects. The method may further comprise selecting a means for dissociating the sample objects from the plurality of means for dissociating the sample objects based on the sample type. For example, if the sample type indicates that the sample is a body fluid sample and/or if the sample type indicates that the sample is a body fluid sample, the sample objects may be dissociated by chemical dissociation (e.g. enzymatic digestion), whereas if the sample type indicates that the sample is a biopsy sample, if the sample type indicates that the sample is a resection sample and/or if the sample type indicates that the sample is an aspiration sample the sample objects may be dissociated by mechanical means, e.g. in addition to chemical dissociation/enzymatic digestion.

[0055] Preferably, a plurality of dissociation protocols are provided, e.g. on a storage medium such as a storage medium of the controller of the system according to the invention described below. One of said dissociation protocols may be selected based on the sample type. Each dissociation protocol may specify a number of dissociation steps that are to be performed and/or means for dissociation sample objects that are to be used. The plurality of dissociation protocols may for example be provided as a library, in particular as an extendable library, to which a user may e.g. add further dissociation protocols.

[0056] In some embodiments, the method may comprise performing one or more control measurements on some or all of the batches of sample objects prior to performing the (main) measurement on the sample objects of the respective batch in the measurement volume. The one or more control measurements may for example comprise determining an object count and/or a cell count (e.g. a total number or density of sample objects and cells, respectively, in the respective batch) and/or a cell viability measure (e.g. a percentage of viable cells in the respective batch) for one or more of the batches of sample objects. The control measurements may for example be performed by optical means (e.g. using an optical sensor such as a photodiode or a camera), by electrical means (e.g. using one or more electrodes to determine a conductivity, an impedance and/or a permittivity) and/or by mechanical means (e.g. for sorting cells by one or more characteristics such as viability).

[0057] The method may comprise manipulating the batches based on a result of the control measurements prior to performing the main measurements in the measurement volume. In particular, a density of sample objects of the respective batch may be adjusted based on the determined object count by supplying a carrier or measurement fluid, e.g. through the measurement fluid inlet of the microfluidic device. For example, fluid may be added if the object density of a batch is above a certain threshold or a variable amount of fluid may be added depending on the object density. This may ensure that the density of sample objects in the batch is sufficiently small such that different sample objects can be analyzed separately.

[0058] The first measurement and the second measurement as well as any additional measurement on sample objects in the measurement volume are not particularly limited to any detection or measurement technique, but may be any type of measurement suitable for analyzing single cells and/or cell aggregates. The measurements may for example be or may comprise an electrical measurement (e.g. a measurement for determining a conductivity, an impedance and/or a permittivity) and/or an optical measurement (e.g. a measurement for determining an intensity of light transmitted, scattered, reflected and/or emitted by the sample objects).

[0059] Preferably, the measurements correspond to or comprise taking one or more microscopic

images, in particular phase shift images of sample objects in the measurement volume. In other words, performing the first measurement and/or performing the second measurement may comprise or correspond to taking an image of the sample objects in the measurement volume using a microscope, in particular taking a phase shift image of the sample objects in the measurement volume using a phase-contrast microscope, in particular a quantitative phase-contrast microscope, for example a digital holographic microscope. As used herein, a phase shift image may refer to an image that encodes a phase shift of light (e.g. an absolute phase shift or a phase shift modulo 2π) at one or more wavelengths as a function of position. The phase shift may for example be the phase shift of light that is transmitted through or reflected off the sample objects in the measurement volume.

[0060] Additionally or alternatively, different microscopy techniques such as bright-field microscopy, fluorescence microscopy, and/or super-resolution microscopy may be used, e.g. to obtain an amplitude or intensity image. In some examples, the measurements may also be or comprise one or more spectroscopic measurements on the sample objects in the measurement volume, wherein a spectroscopic measurement may for example comprise recording one or more optical spectra of one or more sample objects, e.g. an absorption spectrum and/or an emission spectrum in one or more wavelength ranges.

[0061] Performing the first measurement, the second measurement and/or any additional measurement may comprise hydrodynamically and/or viscoelastically focusing the respective sample objects in the measurement volume, for example in a focal plane of a microscope. Hydrodynamic focusing may comprise generating one or more sheath flows, in particular laminar sheath flows, surrounding a carrier or measurement fluid flow in the measurement volume, e.g. to confine sample objects in the carrier or measurement fluid flow along one or more directions to a smaller volume, for example at the center of the measurement volume or in the vicinity of a sidewall of the measurement volume. Viscoelastic focusing may for example be achieved by providing a viscoelastic carrier or measurement fluid, e.g. a fluid containing a shear-thinning polymer such as poly(ethylene oxide) (PEO) and/or poly(vinyl pyrrolidone) (PVP). A viscoelastic fluid may exert hydrodynamic forces on the sample objects, thereby inducing a controlled motion perpendicular to a direction of flow.

[0062] In some embodiments, the method may also comprise performing an analysis of sample objects based on the first measurement, the second measurement and/or any additional measurements. This may for example comprise identifying sample objects and/or their constituents (e.g. individual cells) in a microscopic image, e.g. using classical and/or artificial intelligence-based (AI-based) computer vision techniques such as image segmentation algorithms and/or neural network-based classifiers. This may also comprise determining one or more parameters, in particular morphological parameters, of sample objects and/or their constituents in a microscopic image such as a size (e.g. an equivalent diameter), an optical height (e.g. a mean phase shift and/or a maximum phase shift), an aspect ratio, a number of cells in the respective sample object and/or a type of the cells in the respective sample object. This may for example allow for obtaining a distribution of these parameters over the sample objects and/or cells of a batch as well as for performing single-cell phenotyping of the cells contained in the sample objects. The results of the analysis of the sample objects may be used for diagnostic, prognostic and/or therapeutic purposes, for example to diagnose a medical condition, to assess a severity of a medical condition such as the malignancy of a cancer and/or to select a certain therapy such as a certain drug. In some embodiments, the measurements may be used to probe a stiffness and/or an elasticity of the sample objects, e.g. as described in D. R. Gossett, Proc. Natl. Acad. Sci. 109(20), 7630-7635 (2012). A higher stiffness may indicate malignancy in some examples.

[0063] The invention further provides a system for analyzing a sample comprising one or more sample objects containing biological cells, the one or more sample objects comprising one or more single cells, one or more cell aggregates and/or one or more pieces of continuous tissue material.

The system is configured for use with a microfluidic device according to any one of the embodiments described herein. The system comprises a microfluidics unit configured to generate a flow of a carrier fluid through the insertion volume of the microfluidic device to transport sample objects from the insertion volume to the sorting unit of the microfluidic device. The system further comprises a measurement device configured to perform measurements on single cells and/or cell aggregates in the measurement volume of the microfluidic device. The system also comprises a controller configured to control the microfluidics unit, the measurement device, the sorting unit of the microfluidic device and/or the dissociation unit of the microfluidic device to execute a method according to any one of the embodiments described herein.

[0064] The system may be configured to receive a microfluidic device according to any one of the embodiments described herein. The system may for example comprise a mount that is configured to receive the microfluidic device. The mount may further be configured to provide fluid, electrical and/or optical connections to the microfluidic device, e.g. to connect the microfluidic device to the microfluidics unit and/or to the controller. In some embodiments, the microfluidic device or parts thereof may be integrated into the system.

[0065] The microfluidics unit may comprise one or more pumps and/or one or more valves for generating and controlling fluid flows within the microfluidic device. Additionally or alternatively, the microfluidics unit may be configured to control one or more pumps and/or one or more valves integrated into the microfluidic device for generating and controlling fluid flows within the microfluidic device, e.g. via corresponding electrical and/or fluid connections. The microfluidics unit may further comprise one or more reservoirs for storing fluids such as the carrier fluid, one or more cleaning fluids, one or more dissolving or enzymatic solutions and/or a measurement fluid and for supplying said fluids to the microfluidic device. The microfluidics unit may for example comprise a reservoir storing a cleaning fluid, wherein the cleaning fluid may e.g. comprise a chemical substance such as hypochloride and/or a cleaning enzyme for cleaning the microfluidic device or a part thereof. The microfluidics unit may be configured to provide the cleaning solution to the microfluidic device or said part thereof, e.g. to flush the microfluidic device or said part thereof with the cleaning solution. In some embodiments, the reservoir for storing the cleaning fluid may also be comprised in the microfluidic device itself.

[0066] The measurement device may for example be configured to perform electrical and/or optical measurements on single cells and/or cell aggregates in the measurement volume. The measurement device may for example be or comprise a microscope configured to take microscopic images of single cells and/or cell aggregates in the measurement volume.

[0067] In a preferred embodiment, the measurement device is or comprises a phase-contrast microscope, in particular a quantitative phase-contrast microscope such as a digital holographic microscope configured to take phase shift images of single cells and/or cell aggregates in the measurement volume of the microfluidic device. The microscope may for example be configured to obtain the phase shift through interference of light, e.g. between a probe or imaging beam and a reference beam. In other examples, the quantitative phase-contrast microscope may be a ptychographic imaging device that is configured to perform ptychographic imaging without a reference beam, e.g. by recording interference patterns without a reference phase. The quantitative phase-contrast microscope may be configured to take one or more interference images and to reconstruct a phase shift image from the one or interference images.

[0068] Preferably, the quantitative phase-contrast microscope is a digital holographic microscope, which is configured to take quantitative phase shift images as well as amplitude or intensity images, wherein an intensity image may encode an intensity of light as a function of position, e.g. an intensity of light reflected off or transmitted through an imaging sample such as the sample objects in the measurement volume as a function of the position in the imaging sample. The digital holographic microscope may for example be configured to interfere an image of the imaging sample, e.g. an imaging beam of light transmitted through the imaging sample, with a reference

beam, wherein the reference beam may pass through the imaging sample or may not pass through the imaging sample. The digital holographic microscope may be configured to extract or reconstruct the phase shift and intensity images from one or more interference images, e.g. by reconstructing a wave front of light transmitted through or reflected off the imaging sample. The digital holographic microscope may be an on-axis digital holographic microscope, in which the imaging beam and the reference beam propagate along the same axis when interfering. Preferably, the digital holographic microscope is an off-axis digital holographic microscope, in which the imaging beam and the reference beam interfere under an angle and which may be configured to extract a phase shift image from a single interference image of the imaging sample. Such digital holographic microscopes are for example known from EP 1 524 491 A1 and EP 2 357 539 A1.

[0069] The controller may be implemented in hardware, software or a combination thereof. The controller may for example comprise a processing device and a storage medium, wherein the storage medium contains instructions for execution by the processing device to provide the functionality described herein. The storage medium may further be configured to store data pertaining to the measurements performed by the measurement device (e.g. images and/or data extracted from the images). In some embodiments, the controller and/or the system may additionally or alternatively comprise a second storage medium for storing said data. The system may further comprise a communication module for transferring said data to an external device, for example via a network connection and/or via the internet.

[0070] The system may also comprise one or more means for dissociating sample objects into single cells and/or cell aggregates at least in part. The system may for example comprise an acoustic source, in particular an ultrasound source, that is configured to apply acoustic waves (e.g. ultrasound) to sample objects in the dissociation unit of the measurement device. The controller may be configured to control the acoustic source to dissociate sample objects in the dissociation unit by applying acoustic waves to the sample objects.

[0071] In some embodiments, the microfluidics unit may be configured to provide a dissolving solution (e.g. an enzymatic solution) to the dissociation unit of the microfluidic device, the dissolving solution comprising one or more chemical and/or biochemical substances for chemically dissolving sample objects, for example by enzymatic digestion, e.g. as detailed above. The controller may be configured to control the microfluidics unit to dissociate sample objects in the dissociation unit by supplying the dissolving solution to the dissociation unit.

[0072] In some embodiments, the system may comprise a stirrer, particularly a magnetic stirrer, that is configured to move a stir bar, particularly a magnetic stir bar, in the dissociation unit of the microfluidic device, e.g. as described above. The magnetic stirrer may for example be arranged on or in the mount for the microfluidic device.

[0073] In some examples, the device may also comprise one or more microfluidic devices according to any one of the embodiments described herein. The microfluidic device may be a removable component of the system according to the invention (e.g. such that the microfluidic device can be exchanged against a different microfluidic device) or may be permanently integrated into the system. Additionally or alternatively, one or more of the components of the microfluidic device according to the invention described herein may be part of the system according to the invention instead of being integrated into the microfluidic device. For example, the dissociation unit (or a part thereof) may not be part of the microfluidic device, but may be comprised in the system instead, wherein the dissociation unit may e.g. be configured to be connected to the microfluidic device via a corresponding inlet and outlet of the microfluidic device. Some or all of the components of the microfluidic device that are integrated into the system may be replaceable, e.g. at least the insertion volume, the sorting unit and/or the dissociation unit, and may for example be embodied as modules that can be replaced individually, e.g. as described above for the microfluidic device according to the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0074] In the following, a detailed description of the invention and exemplary embodiments thereof is given with reference to the figures. The figures show schematic illustrations of:

[0075] FIG. **1** is a microfluidic device for analyzing a sample comprising one or more sample objects containing biological cells according to an exemplary embodiment of the invention;

[0076] FIG. **2a** is a microfluidic device comprising a plurality of dissociation volumes in accordance with an exemplary embodiment of the invention;

[0077] FIG. **2b** is a microfluidic device comprising a plurality of dissociation volumes in accordance with another exemplary embodiment of the invention;

[0078] FIG. **3** is a microfluidic device comprising a sample collection unit and a preparatory unit according to an exemplary embodiment of the invention;

[0079] FIG. **4** sample collection units for a microfluidic device in accordance with exemplary embodiments of the invention;

[0080] FIG. **5** the insertion of samples into the sample collection units of FIG. **4** according to exemplary embodiments of the invention;

[0081] FIG. **6** is a system for analyzing a sample comprising one or more sample objects containing biological cells according to an exemplary embodiment of the invention;

[0082] FIG. **7** is a flow chart of a method of analyzing a sample comprising one or more sample objects containing biological cells according to an exemplary embodiment of the invention; and

[0083] FIG. **8** experimental data demonstrating the sorting and dissociation of sample objects containing biological cells.

DETAILED DESCRIPTION

[0084] FIG. **1** shows a schematic illustration (not to scale) of a microfluidic device **100** for analyzing a sample **102** comprising sample objects **102A-C** containing biological cells according to an exemplary embodiment of the invention. The sample **102** comprises a plurality of single cells **102A**, a plurality of cell aggregates **102B** as well as one or more pieces of continuous tissue material **102C**. The sample **102** may for example be a biopsy or aspiration sample obtained from a patient, e.g. from cancer tissue in an organ of the patient. In other examples, the sample **102** may for example be a body fluid sample such as a blood sample or a smear sample such as an oral smear.

[0085] The microfluidic device **100** comprises a substrate **104**, which may for example comprise or consist of a transparent thermoplastic such as polymethyl methacrylate (PMMA). An insertion volume **106** is formed in the substrate **104**, wherein the insertion volume **106** is configured to receive the sample **102**, e.g. through an inlet or opening **104A**. A width, a height and a length of the insertion volume **106** may for example be between 0.5 mm and 50 mm, in one example between 1 mm and 10 mm. A volume of the insertion volume may for example be between 1 mm³ and 20 cm³. The insertion volume **106** may be part of a sample collection unit as detailed below with reference to FIGS. **4** and **5**.

[0086] The microfluidic device **100** further comprises a sorting unit **108** that is configured to sort the sample objects **102A-C** by size, weight and/or stiffness. An inlet of the sorting unit **108** is in fluid communication with the insertion volume **106** via a microfluidic channel. A first outlet **108A** of the sorting unit **108** is in fluid communication with a measurement volume **110** formed in the substrate **104** via another microfluidic channel. A second outlet **108B** of the sorting unit **108** is in fluid communication with a dissociation unit **114** via yet another microfluidic channel.

[0087] The sorting unit **108** comprises a filter **108C**, which may for example be an inertia-based cell filter, an acoustic cell filter, a hydrodynamic cell filter, a dead-end cell filter or a cross-flow cell filter. In some embodiments, the sorting unit **108** may comprise a plurality of filters selected from

the aforementioned types of filters, which may for example be arranged in series. The filter **108C** is configured to direct sample objects having a smaller size, a smaller weight and/or a lower stiffness from the inlet towards the first outlet **108A**. The filter **108C** is further configured to direct sample objects having a larger size, a larger weight and/or a higher stiffness from the inlet towards the second outlet **108B**. The filter **108C** may for example be configured to direct sample objects having a size smaller than a (first) sorting threshold towards the first outlet **108A** and sample objects having a size larger than the sorting threshold towards the second outlet **108B**. The sorting threshold may for example be between 5 μm and 1000 μm , in some examples between 10 μm and 100 μm , e.g. 40 μm . This may for example allow for separating single cells **102A** and smaller cell aggregates **102B** from larger cell aggregates **102B** and pieces of tissue **102C**. The single cells **102A** and the smaller cell aggregates **102B** may pass through the filter **108C** to the measurement volume **110**, while the larger cell aggregates **102B** and pieces of tissue **102C** may be diverted to the dissociation unit **114**. The sorting threshold may for example be chosen based on the physical dimensions of the measurement volume **110** (e.g. to prevent clogging thereof) and/or based on hydrodynamic considerations, for example to facilitate a hydrodynamic and/or viscoelastic focusing of sample objects in the measurement volume **110**. In some embodiments, the microfluidic device **100** may further comprise another filter (not shown) and/or another sorter (not shown) between the outlet of the dissociation unit **114** and the inlet of the measurement volume **110**, e.g. as described below with reference to FIGS. **2a**, **2b**.

[0088] The measurement volume **110** is configured for performing measurements on sample objects contained therein, for example electrical and/or optical measurements. In one example, the measurement volume **110** is a microfluidic channel, which may for example have a width between 10 μm and 1000 μm , in one example between 30 μm and 500 μm , and a height between 10 μm and 1000 μm , in some examples between 30 μm and 500 μm , in one example between 30 μm and 100 μm . The measurement volume **110** may for example comprise an optical detection window **112**, wherein the detection window **112** is configured for performing microscopic imaging and in particular phase-contrast imaging therethrough. A transmitted wavefront error of the detection window **112** may for example be no more than $\lambda/2$. Additionally or alternatively, the microfluidic device **100** may for example comprise one or more electrodes for performing electrical measurements in the measurement volume **110**. The measurement volume **110** may be in fluid communication with an outlet or opening **104B** and/or with a waste reservoir (not shown) of the microfluidic device **100**.

[0089] The dissociation unit **114** comprises a dissociation volume **114A**, wherein an inlet of the dissociation volume **114A** is in fluid communication with the second outlet **108B** of the sorting unit **108** and an outlet of the dissociation volume **114A** is in fluid communication with the measurement volume **110** as illustrated in FIG. **1**. The dissociation unit **114** comprises means for dissociating sample objects into smaller objects such as single cells and/or cell aggregates at least in part. In the example of FIG. **1**, the means for dissociating sample objects comprise a reservoir **116** that is configured to store a dissolving solution, in particular an enzymatic solution, and to supply said solution to the dissociation volume **114A**. The enzymatic solution may comprise one or more enzymes for dissolving sample objects by enzymatic digestion, for example trypsin and/or collagenase. The sample objects may be exposed to the enzymatic solution in the dissociation volume **114A** and may thus be dissociated into smaller objects prior to being transferred to the measurement volume **110**.

[0090] FIG. **2a** shows a schematic illustration (not to scale) of a microfluidic device **200A** according to another exemplary embodiment of the invention. The microfluidic device **200A** is similar to the microfluidic device **100** of FIG. **1** and also comprises a sorting unit **108**, a dissociation unit **114**, an insertion volume (not shown) upstream of the sorting unit **108** and the dissociation unit **114** (i.e. to the left of the arrangement shown in FIG. **2**) as well as a measurement volume (not shown) downstream of the sorting unit **108** and the dissociation unit **114** (i.e. to the

right of the arrangement shown in FIG. 2).

[0091] In the example of FIG. 2a, the dissociation unit **114** comprises a plurality of dissociation volumes or dissociation channels, namely a first dissociation volume **114A** and a second dissociation volume **114B**, and a plurality of means for dissociating sample objects. The dissociation volumes **114A-B** are arranged in parallel with each dissociation volume **114A-B** being associated or equipped with a different means for dissociating sample objects. Furthermore, each dissociation volume **114A-B** is associated or equipped with a means **118A-B** for controlling a fluid flow through the respective dissociation volume **114A-B**. Said means **118A-B** may for example be or comprise a pump and/or a valve, e.g. such that sample objects may be selectively directed into one of the dissociation volumes **114A-B**.

[0092] The means for dissociating sample objects comprise a reservoir **116** for supplying a dissolving solution, for example an enzymatic solution, which is arranged upstream of each of the dissociation volumes **114A-B**, e.g. such that the dissolving/enzymatic solution can be supplied to each of the dissociation volumes **114A-B** simultaneously. In some embodiments, the reservoir **116** may comprise a plurality of sub-reservoirs, each of which may for example be configured to supply a different dissolving or enzymatic solution, e.g. a solution containing a different substance/enzyme or a different combination of substances/enzymes.

[0093] The means for dissociating sample objects further comprise a grid or array of static obstacles **116A** arranged in the first dissociation volume **114A**, wherein the static obstacles **116A** are configured to mechanically dissociate sample objects colliding with one of the obstacles **116A**. Each of the static obstacles **116A** may for example be or may comprise a blade that is configured to cut or slice through a sample object upon impact.

[0094] The means for dissociating sample objects further comprises a plurality of constricted or narrower portions **116B** that are arranged along the second dissociation volume **114B** in between a plurality of non-constricted or wider portions. Physical dimensions of the constricted portions **116B** (e.g. a width in the drawing plane of FIG. 2a and/or a height perpendicular to the drawing plane of FIG. 2a) and/or a radius of curvature of the sidewalls of the second dissociation volume **114A** are chosen such that strong shear forces are generated on sample objects flowing through the second dissociation volume **114B** that are sufficiently large to break sample objects apart, e.g. to separate cells of a loosely-bound cell aggregate. In some embodiments, a cross-sectional area, a width and/or a height of the constricted portions may decrease successively along the second dissociation volume **114B** (not shown).

[0095] The microfluidic device **200A** comprises a sorter **202** that is arranged at the outlet of the dissociation unit **114**. The sorter **202** may be similar or identical to the sorting unit **108** and is also configured to sort sample objects by size, weight and/or stiffness by directing sample objects having a smaller size (e.g., a size below a second sorting threshold, which may be equal to the first sorting threshold of the sorting unit **108**), a smaller weight and/or a lower stiffness towards a first outlet **202A** that is in fluid communication with the measurement volume and by directing sample objects having a larger size (e.g. a size above the second sorting threshold), a larger weight and/or a higher stiffness towards a second outlet **202B**. The second outlet **202B** is in fluid communication with an inlet of the dissociation unit **114**, e.g. such that larger sample objects may cycle through the dissociation unit **114** repeatedly for sequentially dissociating larger sample objects. In other embodiments, the outlet of the dissociation unit **114** may be in fluid communication with the measurement volume via the sorting unit **108** (i.e. the sorting unit **108** may also provide the functionality of the sorter **202**), e.g. by connecting the outlet of the dissociation unit **114** with an inlet upstream of the sorting unit **108**, for example as detailed below with reference to FIG. 2b.

[0096] The microfluidic device **200A** also comprises one or more filters **204** that are arranged upstream of the measurement volume between the first outlet **108A** of the sorting unit **108** and the outlet of the dissociation unit **114**, respectively, and the inlet of the measurement volume. The filters **204** may for example comprise one or more mechanical filters such as a dead-end cell filter

and/or a cross-flow cell filter, e.g. to prevent sample objects larger than a filtering threshold, which may for example have passed through the sorting unit **108** or the sorter **202** accidentally, from entering the measurement volume. The filtering threshold may be equal to the first and/or second sorting thresholds. In other examples, the filtering threshold may be larger than the first and/or second sorting thresholds. In some embodiments, the filters **204** may comprise a cell-stiffness sorter that is configured to filter or sort sample objects by stiffness, e.g. to filter out dead and/or non-viable cells upstream of the measurement volume.

[0097] FIG. **2b** shows a schematic illustration (not to scale) of a microfluidic device **200B** according to yet another exemplary embodiment of the invention. The microfluidic device **200B** is similar to the microfluidic device **200A** of FIG. **2a** and also comprises a sorting unit **108**, a dissociation unit **114**, an insertion volume (not shown) upstream of the sorting unit **108** and the dissociation unit **114** (i.e. to the left of the arrangement shown in FIG. **2**) as well as a measurement volume (not shown) downstream of the sorting unit **108** and the dissociation unit **114** (i.e. to the right of the arrangement shown in FIG. **2**).

[0098] Similar to the microfluidic device **200A**, the dissociation unit **114** comprises a plurality of dissociation volumes or dissociation channels, namely a first dissociation volume **114C** and a second dissociation volume **114D**, and a plurality of means for dissociating sample objects. The means for dissociating sample objects comprise a reservoir **116** for supplying a dissolving solution, e.g. as detailed above for the microfluidic device **200A**.

[0099] The means for dissociating sample objects further comprise a magnetic stirrer **116C** that is configured to rotate a magnetic stir bar in the first dissociation volume **114C** for dissociating sample objects. The magnetic stir bar may for example also comprise one or more blades configured to cut or slice through a sample object. The means for dissociating sample objects also comprise an acoustic emitter, in this case an ultrasound emitter **116D**, arranged in a side wall of the second dissociation volume **114D**, wherein the ultrasound emitter **116D** is configured to apply ultrasound to sample objects in the second dissociation volume **114D**. The ultrasound emitter **116D** may for example be an active emitter comprising a piezo element, a capacitive element and/or an inductive element that is configured to convert an electrical signal applied to the ultrasound emitter to ultrasound. In other examples, the ultrasound emitter **116D** may be a passive emitter, which may for example be configured to couple ultrasound generated by an external source such as a sonotrode into the second dissociation volume **114D**, e.g. as described below for the device **600** of FIG. **6**.

[0100] In the example of FIG. **2b**, the microfluidic device **200B** does not comprise an additional sorter such as the sorter **202** of the microfluidic device **200A**. Instead, the outlet of the dissociation unit **114** is in fluid communication with the measurement volume via the sorting unit **108** (i.e. the sorting unit **108** also provides the functionality of the sorter **202**). For this, the outlet of the dissociation unit **114** is connected to an inlet upstream of the sorting unit **108**. Sample objects dissociated in the dissociation unit **114** thus have to pass through the sorting unit **108** again before entering the measurement volume.

[0101] In some embodiments, the microfluidic device **200A** and/or the microfluidic device **200B** may comprise more than two dissociation volumes and/or further means for dissociating sample objects, for example some or all of the dissociation volumes **114A-D** and/or some or all of the means **116**, **116A-D** for dissociating sample objects as described above.

[0102] FIG. **3** shows a schematic illustration (not to scale) of a microfluidic device **300** according to yet another exemplary embodiment of the invention. The microfluidic device **300** is similar to the microfluidic devices **100**, **200A**, **200B** of FIGS. **1**, **2a** and **2b**, respectively, and comprises a sample collection unit **302** with an insertion volume for receiving a sample, a dissociation and filtration unit with a sorting unit **108** and a dissociation unit **114** downstream of the sample collection unit **302** as well as a measurement volume **110** downstream of the dissociation and filtration unit. In the measurement volume **110**, measurements may be performed with a

measurement device **608** such as a digital holographic microscope, for example using the system **600** of FIG. **6** described below.

[0103] The sample collection unit **302** comprises a reservoir for supplying a buffer or carrier fluid such as phosphate-buffered saline (PBS) and a flow pump configured to generate a flow of the carrier fluid through the insertion volume to transport sample objects from the insertion volume to the sorting unit **108** and further on to the dissociation unit **114** and the measurement volume **110**. The sample collection unit **302** may for example be embodied as detailed below with reference to FIGS. **4** and **5**.

[0104] The dissociation and filtration unit with the sorting unit **108** and the dissociation unit **114** may for example be embodied as in the microfluidic devices **200A**, **200B** of FIGS. **2a**, **2b**. The dissociation and filtration unit **108/114** may comprise additional components such as a heater, a cooler and/or a carbon-dioxide supply. The heater may for example comprise one or more heating elements and may be configured to heat some or all of the dissociation volumes and/or the reservoir **116**, e.g. to adjust a temperature during the chemical dissociation/enzymatic digestion of the sample objects. The cooler may for example comprise one or more cooling elements (e.g. a thermoelectric cooling element such as a Peltier element) and may e.g. be configured to cool the reservoir **116**, for example for cooling an enzymatic solution contained therein while the device **300** is not in use. The carbon-dioxide supply may be configured to supply carbon-dioxide to the dissociation unit **114**, e.g. to some or all of the dissociation volumes and/or to the reservoir **116**, e.g. to adjust a carbon dioxide concentration during the chemical dissociation/enzymatic digestion of the sample objects.

[0105] In addition to the reservoir **116** for dissolving/enzymatic solutions, the dissociation and filtration unit **108/114** may comprise one or more additional reservoirs, e.g. to supply a buffer or carrier fluid, such as PBS, to supply a cleaning solution for cleaning the device **300** or a part thereof and/or to supply a measurement fluid, such as a viscoelastic fluid (e.g. PBS containing a shear-thinning polymer such as poly(ethylene oxide) (PEO)). The dissociation and filtration unit **108/114** may further comprise a mixing unit, wherein the mixing unit may for example be configured to mix fluids from the reservoir **116** and/or from the additional reservoirs.

[0106] The microfluidic device **300** further comprises a control unit **304** arranged between the dissociation and filtration unit **108/114** and the measurement volume **110**. The control unit **304** may for example be used to perform control measurements, e.g. optical and/or electrical control measurements, on the sample objects, for example to determine an object count (e.g. the number or density of sample objects and/or cells in a given batch) and/or a cell viability measure.

[0107] The microfluidic device **300** further comprises a preparatory unit **306** arranged between the dissociation and filtration unit **108/114** and the measurement volume **110**. The preparatory unit **306** may for example comprise one or more filters similar to the sorting unit **108** and/or the filters **204** of the microfluidic devices **200A**, **200B**, e.g. a first filter **204A** such as a cross-flow cell filter for directing larger sample objects to a waste reservoir and/or for exchanging the carrier fluid and/or the dissolving solution (e.g. by a measurement fluid) and a second filter **204B** such as a dead-end cell filter downstream of the first filter **204A** to prevent clogging of the measurement volume as illustrated in FIG. **3**. The preparatory unit **306** may further comprise a measurement fluid inlet for supplying a measurement fluid, e.g. in addition to the carrier fluid and/or the dissolving/enzymatic solution and/or to replace the carrier fluid and/or the dissolving/enzymatic solution prior to performing measurements on the sample objects (for example to adjust a density of sample objects and/or to provide a viscoelastic substance).

[0108] The microfluidic device **300** also comprises one or more sheath fluid inlets arranged upstream of and/or in the measurement volume. The sheath fluid inlets may be configured to generate one or more sheath flows (e.g. a pair of sheath flows for confinement along a single direction or two pairs of sheath flows for confinement along two orthogonal directions) for hydrodynamic focusing of the sample objects in the measurement volume.

[0109] The microfluidic device **300** further comprises a post-processing unit **308** downstream of

the measurement volume **110**. The post-processing unit **308** may for example comprise one or more outlets similar to the outlet **104B** of the microfluidic device **100** of FIG. **1** and/or one or more reservoirs for collecting sample objects after having performed a measurement thereon in the measurement volume **110**, e.g. for further analyses. In some embodiments, the post-processing unit **308** may also comprise a sorter, which may for example be configured to sort the sample objects based on a measurement performed in the measurement volume **110**, e.g. for sorting the sample objects by cell type. Additionally or alternatively, the post-processing unit **308** may for example comprise a filter for exchanging the carrier fluid or measurement fluid, e.g. by a cell culture medium.

[0110] In some embodiments, some or all of the components of the microfluidic device **300** described above may not be part of the microfluidic device **300**, but may instead be integrated into a system for analyzing a sample comprising one or more sample objects containing biological cells according to the invention, e.g. the system **600** of FIG. **6** described below.

[0111] FIG. **4** depicts schematic illustrations (not to scale) of sample collection units for a microfluidic device such as the microfluidic devices **100**, **200A**, **200B** and **300** in accordance with exemplary embodiments of the invention. One or more of these sample collections units may be permanently integrated into the microfluidic device. Additionally or alternatively, the one or more of these sample collection units may be configured to be removably connected to the microfluidic device, e.g. as single-use plug-in units.

[0112] Panel (a) of FIG. **4** depicts a sample collection unit configured for receiving liquid samples such as a body fluid sample. The sample collection unit comprises a connector that is configured to be coupled to a sample collection device such as a syringe and/or a tube for transferring a sample (e.g. a sample fluid) contained in the sample collection device to the insertion volume of the microfluidic device. The sample collection unit further comprises a flow pump for generating a flow through the insertion volume to transport sample objects from the insertion volume to the sorting unit.

[0113] Panel (b) of FIG. **4** depicts a sample collection unit configured for receiving solid samples such as a biopsy sample and/or an aspiration sample. The sample collection unit comprises an opening that is configured to receive a needle such as an aspiration needle (e.g. a fine-needle aspiration (FNA) needle) and/or a biopsy needle (e.g. a fine-needle biopsy (FNB) needle). The opening is in fluid communication with the insertion volume for transferring a sample from the needle to the insertion volume. The sample collection unit further comprises a wiper for wiping off the sample from the needle as illustrated in panel (a) of FIG. **5** (left plot). The wiper may for example be arranged on a side wall of the insertion volume as shown in FIG. **4**. The sample collection unit further comprises a flow pump that is configured to generate a flow through the insertion volume to transport sample objects from the insertion volume to the sorting unit as illustrated in panel (a) of FIG. **5** (right plot). The flow pump is further configured to detach a sample from the needle and/or from the wiper by flushing the insertion volume as illustrated in panel (a) of FIG. **5** (center plot).

[0114] Panel (c) of FIG. **4** depicts another example of a sample collection unit that is configured for receiving liquid as well as solid samples from a sample collection device such as a syringe, a spatula or a needle. The sample collection unit comprises a collection tube such as a standard centrifuge tube and a transfer pump for transferring a sample from the collection tube to the insertion volume. The transfer pump may for example be mounted on or in a movable head, wherein the moveable head may be configured to insert an inlet of the transfer pump into the collection tube and into the insertion volume or into an inlet in fluid communication with the insertion volume. The transfer pump may be configured to suck the sample from the collection tube and to release the sample into the insertion volume, cf. panel (b) of FIG. **5** (right plot). The transfer pump may also be configured to flush the collection tube, e.g. for detaching a sample from a sample collection device, cf. panel (b) of FIG. **5** (left plot). In some embodiments, the collection

tube may comprise a wiper for wiping of a sample from a sample collection device, cf. panel (b) of FIG. 5 (center plot).

[0115] In some embodiments, the sample collection unit may comprise two or more of the means for receiving a sample illustrated in panels (a), (b) and (c) of FIG. 4, e.g. the connector of panel (a) as well as the needle opening of panel (b) and/or the collection tube of panel (c).

[0116] FIG. 6 shows a schematic illustration (not to scale) of a system **600** for analyzing a sample comprising one or more sample objects containing biological cells according to an exemplary embodiment of the invention. The system **600** is configured for use with a microfluidic device **602** according to any one of the embodiments described herein, for example any one of the microfluidic devices **100**, **200A**, **200B** and **300** described above. In some embodiments, the system may comprise the microfluidic device **600**.

[0117] The system **600** comprises a mount **604** that is configured to receive the microfluidic device **602** and to hold the microfluidic device **602** in place. The mount **604** may further be configured to provide fluid, electrical and/or optical connections (not shown) to the microfluidic device **602**.

[0118] The system **600** further comprises a microfluidics unit **606** that is configured to generate fluid flows within the microfluidic device **602**, in particular to generate a flow of a carrier fluid through the insertion volume **106** of the microfluidic device **602** to transport sample objects from the insertion volume **106** to the sorting unit (not shown) of the microfluidic device **602** and further on to the dissociation unit **114** and the measurement volume **110** of the microfluidic device **602**. To generate and control the fluid flows, the microfluidic device **602** may comprise one or more pumps, one or more valves and/or one or more reservoirs for storing fluids (not shown). Some or all of the pumps, valves and/or reservoirs of the microfluidic devices **100**, **200A**, **200B** and **300** described above may be part of the microfluidics unit **606** instead of being integrated into the respective microfluidic device. The microfluidics unit **606** may for example comprise a reservoir and/or a pump (not shown) for providing a buffer or carrier fluid to the inlet **104A** of the microfluidic device **602**, e.g. similar to the sample collection unit **302** of the microfluidic device **300**. Additionally or alternatively, the microfluidics unit **606** may comprise a reservoir and/or a pump (not shown) for supplying a dissolving solution, for example an enzymatic solution, to the microfluidic device **602** via an inlet **104C**, e.g. to the reservoir **116** of the microfluidic devices **100**, **200A**, **200B** and/or directly to a dissociation volume such as the dissociation volume **114A** of the microfluidic device **100**.

[0119] The system **600** comprises a measurement device **608** configured to perform measurements on single cells and/or cell aggregates in the measurement volume **110** of the microfluidic device **602**. In the example of FIG. 6, the measurement device **608** is a quantitative phase-contrast microscope, namely a digital holographic microscope (DHM) that is configured to take phase shift images as well as amplitude or intensity images of single cells and/or cell aggregates in the measurement volume **110** of the microfluidic device **602**.

[0120] The digital holographic microscope **608** comprises an imaging system with an objective **608A**, a holographic imaging system **608B** and an imaging lens **608C**. The imaging system is configured to image a focal plane of the imaging system onto a camera **608D**, e.g. through a detection window (not shown) of the microfluidic system **602**. The microscope **608** may further comprise an illumination source (not shown) for illuminating the measurement volume **110**.

[0121] The holographic imaging system **608B** is configured to create an interference image on the camera **608D**, e.g. by interfering an imaging beam with a reference beam on the camera **608D**. The imaging beam may for example be a beam that passes through the measurement volume **110** and propagates from the focal plane of the imaging system to the camera **608D** along a first optical path through the holographic imaging system **608B**. The reference beam may for example be a beam propagating to the camera **608D** along a second optical path through the holographic imaging system **608B**. In some examples, the reference beam may be split from the imaging beam, e.g. using a beam splitter or a diffraction grating, i.e. the reference beam may also have passed through

the measurement volume **110** and may propagate to the camera **608D** from the focal plane of the imaging system along the second optical path. In other examples, the reference beam may not have passed through the measurement volume **110** and may e.g. be split from the imaging beam in front of the measurement volume **110**.

[0122] The digital holographic microscope **608** may be an on-axis digital holographic microscope, in which the imaging beam and the reference beam propagate along the same axis when interfering, i.e. interfere at an angle of 0° . The digital holographic microscope **608** may for example be configured to extract or reconstruct a phase shift image as well as an intensity image of the sample from a plurality of interference images, e.g. by varying a phase offset between the reference and imaging beams using the holographic imaging system **608B**. Preferably, the microscope **608** is an off-axis digital holographic microscope, in which the imaging beam and the reference beam interfere under an angle. In this case, the digital holographic microscope **608** may be configured to extract or reconstruct the phase shift image as well as the intensity image of the sample from a single interference image.

[0123] In other embodiments, the measurement device **608** may comprise a different type of microscope, e.g. a bright-field microscope and/or a fluorescence microscope, instead of or in addition to the digital holographic microscope. Additionally or alternatively, the measurement device **608** may also be configured to perform other types of optical measurements, e.g. spectroscopic measurements, and/or electrical measurements such as impedance measurements.

[0124] The system **600** further comprises a controller **610** that is configured to control the microfluidics unit **606**, the measurement device **608**, the sorting unit of the microfluidic device **602** and/or the dissociation unit **114** of the microfluidic device **602**, e.g. by generating corresponding analog and/or digital control signals. The controller **610** may be implemented in hardware, software or a combination thereof. The controller **610** may for example comprise a processing device and a memory or storage medium storing instructions for execution by the processing device to provide the functionality described herein. The controller **306** may for example comprise a central processing unit (CPU), a graphics processing unit (GPU), a field programmable gate array (FPGA), an application-specific integrated circuit (ASIC), and/or a microcontroller.

[0125] The controller **610** is configured to execute a method of analyzing a sample comprising one or more sample objects containing biological cells according to any one of the embodiments described herein by controlling the microfluidics unit **606**, the measurement device **608**, the sorting unit of the microfluidic device **602** and/or the dissociation unit **114** of the microfluidic device **602** accordingly. The controller **610** may in particular be configured to execute some or all of the steps of the method **700** of FIG. 7 described below.

[0126] In some embodiments, the system **600** may comprise one or more means for dissociating sample objects, wherein the controller **610** may be configured to control said means. The system **600** may for example comprise one or more reservoirs for supplying a dissolving solution, for example an enzymatic solution, to the dissociation unit of the microfluidic device **602**, e.g. as described above. The system **600** may also comprise an acoustic source, in this example an ultrasound source **612**, that is configured to apply acoustic waves/ultrasound to sample objects in the dissociation unit of the microfluidic device **602**. The ultrasound source **612** may for example comprise a sonotrode that is configured to apply ultrasound to a passive ultrasound emitter of the microfluidic device **602** such as the ultrasound emitter **116D** of the microfluidic device **200B** of FIG. 2b. The system **600** may also comprise a magnetic stirrer that is configured to move a magnetic stir bar in the microfluidic device, e.g. similar to the magnetic stirrer **116C** of the microfluidic device **200B** of FIG. 2b. In some embodiments, the ultrasound source or a part thereof and/or the magnetic stirrer or a part thereof may be integrated into the mount **602**.

[0127] FIG. 7 depicts a flow chart of a method **700** of analyzing a sample comprising one or more sample objects containing biological cells according to an exemplary embodiment of the invention. The method **700** is executed using a microfluidic device according to any one of the embodiments

described herein. The method **700** may for example be executed using a microfluidic device similar to the microfluidic devices **200A**, **200B** of FIGS. **2a**, **2b** that contains the four dissociation volumes **114A-D** and the means **116**, **116A-D** for dissociating sample objects and the system **600** of FIG. **6**, which are used as examples for illustration purposes in the following. This is, however, not intended to be limiting in any way and the method **700** may also be implemented using a microfluidic device such as the microfluidic device **100**, **300** or **602** and/or may be implemented without the system **600** or with a different system for analyzing a sample comprising one or more sample objects containing biological cell according to any one of the embodiments described herein. Furthermore, the method **700** is not limited to the order of execution shown in the flowchart of FIG. **7**. As far as technically feasible, the method **700** may be executed in an arbitrary order and parts thereof may be executed simultaneously at least in part, e.g. steps **708** and **710**.

[0128] The method **700** may be used for analyzing a wide range of samples comprising sample objects such as single cells **102A**, cell aggregates **102B** and/or one or more pieces of continuous tissue material **102C**, for example biopsy samples, aspiration samples, body fluid samples and/or smear samples.

[0129] The method **700** comprises, in step **702**, determining a sample type of the sample **102**, wherein the sample type characterizes one or more of a procedure used for obtaining the sample **102** from a patient (e.g. biopsy, aspiration, puncture, lavage or smear), a body part that the sample **102** was obtained from (e.g. an organ, a type of tissue or a type of body fluid), a size of the sample objects **102A-C** in the sample **102** and a weight of the sample objects **102A-C** in the sample **102**. The sample type may for example be entered manually by a user and/or may be determined by scanning a label such as a barcode associated with the sample **102** and/or with a sample collection device.

[0130] Based on the sample type, a workflow for analyzing the sample **102** is determined. This may in particular comprise choosing a number of dissociation steps that are to be performed, means for dissociating sample objects that are to be used (e.g. selecting a dissolving solution that is to be used) and/or parameters for the dissociation of the sample objects such as a duration or a concentration of enzymes.

[0131] The sample **102** may be provided in the insertion volume **106** of the microfluidic device **200A/200B** prior to execution of the method **700** or as part of the method **700**, e.g. as illustrated in FIGS. **4** and **5**. This may also comprise providing a buffer or carrier fluid such as PBS in the insertion volume **106** for suspending the sample objects **102A-C** therein. Providing the sample **102** may comprise flushing the insertion volume **106** and/or a collection tube for detaching the sample **102** from a sample collection device, such as a biopsy needle, and/or may comprise wiping off the sample **102** from the sample collection device at a wiper.

[0132] In step **704**, a flow of the carrier fluid is generated through the insertion volume **106** to transport sample objects **102A-C** from the insertion volume **106** to the sorting unit **108**. In step **706**, the sample objects **102A-C** pass through the sorting unit **108**. The sorting unit **108** sorts the sample objects **102A-C** by size, weight and/or stiffness into a first batch of sample objects, which may e.g. comprise some or all of the sample objects **102A-C** having a size smaller than a sorting threshold (e.g. smaller than 40 μm), and a first residual sample, which may contain the remaining sample objects **102A-C** that are not contained in the first batch. The first residual sample may in particular contain the sample objects **102A-C** having a size larger than the sorting threshold.

[0133] The first batch of sample objects is then transferred to the measurement volume **110** via the first outlet **108A** of the sorting unit **108** and a first measurement is performed on the sample objects of the first batch in step **708**. The first measurement may for example comprise taking one or more phase shift images of the sample objects in the first batch using the digital holographic microscope **608** of the system **600**.

[0134] Prior to performing the first measurement, a density of the sample objects **102A-C** in the first batch may be determined (e.g. number of sample objects per volume or a mean inter-object

distance) and may optionally be adjusted, e.g. by adding more fluid, for example to ensure that the density is sufficiently low such that separate sample objects **102A-C** can easily be discerned in the phase shift images. Before entering the measurement volume **110**, the first batch of sample objects **102A-C** may also be filtered using the filters **204**, e.g. to prevent larger sample objects that might clog the measurement volume **110** from entering the measurement volume **110**.

[0135] For taking the phase shift images, the sample objects **102A-C** in the first batch may be focused hydrodynamically and/or viscoelastically in a focal plane of the microscope **608**, for example by generating one or more sheath flows as described above. Preferably, the sample objects **102A-C** are focused viscoelastically in the measurement volume **110** by performing the measurement in a viscoelastic fluid, e.g. by using a viscoelastic carrier fluid, by adding a viscoelastic measurement fluid to the carrier fluid prior to performing the measurement and/or by replacing the carrier fluid with a viscoelastic measurement fluid prior to performing the measurement. The viscoelastic fluid may comprise one or more shear-thinning polymers, which may for example be selected from the group consisting of poly(ethylene oxide) (PEO), poly(vinyl pyrrolidone) (PVP), hyaluronic acid (HA) and polyacrylamide (PAA). Preferably, the shear-thinning polymer is poly (ethylene oxide) (PEO), which may e.g. be dissolved in a buffer such as PBS. The shear-thinning polymer may have a molecular weight between 2 MDa and 10 MDa, preferably between 3 MDa and 6 MDa, most preferably between 3.5 MDa and 4.5 MDa, in one example of 4.0 MDa. A mass fraction of the shear-thinning polymer in the viscoelastic fluid may be less than 0.2%, preferably between 0.03% and 0.12%, most preferably between 0.04% and 0.06%, in one example 0.05%. Such a viscoelastic fluid may allow for reliable and accurate viscoelastic focusing of sample objects of different sizes such as single cells and cell aggregates at low flow velocities, thus reducing mechanical stress on the sample objects, which may damage the sample objects and/or influence the measurement. Furthermore, the inventors have found that such a viscoelastic fluid does not induce a formation of artificial cell aggregates even in blood samples, which may for example be prone to polymer-induced platelet aggregation.

[0136] The first residual sample is transferred to the dissociation unit **114** via the second outlet **108B** of the sorting unit **108** for dissociating the sample objects **102A-C** contained therein into single cells and/or cell aggregates at least in part in step **710**. The sample objects **102A-C** may be dissociated using any combination of means for dissociating sample objects described above, for example by chemical dissociation (e.g. enzymatic digestion) and/or mechanical dissociation.

[0137] In some embodiments, the means for dissociating the sample objects **102A-C** in the first residual sample may be selected based on the sample type determined in step **702**. Using the means **118A-D** for controlling fluid flows through the dissociation volumes **114A-D**, the sample objects **102A-C** may be selectively directed to the respective dissociation volume. For example, if the sample is expected to contain larger pieces of tissue **102C** (e.g. if the sample is a biopsy sample), the magnetic stirrer **116C** and/or the blades **116A** may be used for cutting the pieces of tissue **102C** into smaller pieces. On the other hand, if the sample is not expected to contain any larger pieces of tissue **102C** (e.g. if the sample is a body fluid sample), a dissolving solution and/or the constricted portions **116B** may be used to dissociate larger cell aggregates **102B** into single cells and/or smaller cell aggregates. Furthermore, parameters such as an enzyme concentration and/or a duration of the dissociation step (e.g. a duration of exposure to the enzyme solution and/or a duration for which the magnetic stirrer **116C** is activated) may be chosen based on the sample type.

[0138] The method **700** may then return to step **706** for sorting the dissociated sample objects **102A-C** from the first residual sample into a second batch of sample objects and a second residual sample, e.g. using the sorter **202** or the sorting unit **108**, as described above for the first batch of sample objects and the first residual sample. The method **700** may then proceed to step **708** to transfer the second batch of sample objects to measurement volume **110** for performing a second measurement on the dissociated sample objects in the second batch, e.g. as described above for the first batch of sample objects.

[0139] The second residual sample may either be discarded or may be used to obtain one or more additional batches of sample objects by repeatedly performing steps **706** to **710**, i.e. by dissociating the sample objects in a residual sample separated from a previous batch using the dissociation unit **114** and subsequently sorting the dissociated sample objects into the respective additional batch of sample objects and a new residual sample using the sorter **202**. In some embodiments, different means for dissociating sample objects may be used for different dissociation steps. For example, the magnetic stirrer **116C** may be used during the first dissociation step (i.e. for dissociating sample objects **102A-C** in the first residual sample), the blades **116A** may be used during the second dissociation step (i.e. for dissociating sample objects **102A-C** in the second residual sample), the ultrasound emitter **116D** may be used during the third dissociation step (i.e. for dissociating sample objects **102A-C** in the third residual sample) and the constricted portions **116B** may be used during the fourth dissociation step (i.e. for dissociating sample objects **102A-C** in the fourth residual sample). Additionally or alternatively, the sample objects **102A-C** may be exposed to a dissolving solution during some or all of the dissociation steps. In some examples, different dissolving/enzymatic solutions may be used in different dissociations steps, e.g. a first solution in the first dissociation step and a second solution in the second dissociation step.

[0140] The number of dissociations steps that are performed may be determined or chosen based on the sample type determined in step **702**. For example, a single dissociation step may be performed if the sample type indicates that the sample is a body fluid sample and/or if the sample type indicates that the sample is a smear sample, whereas multiple dissociation steps may be performed if the sample type indicates that the sample is a biopsy sample and/or if the sample type indicates that the sample is an aspiration sample.

[0141] FIG. **8** shows experimental data demonstrating the sorting and dissociation of sample objects containing biological cells using a microfluidic device according to the invention. For this, samples comprising squamous cells from an oral smear of a patient were analyzed after sorting and dissociating sample objects using different dissociation protocols (center-right and right plots) and compared to a reference sample for which no sorting or dissociation was performed (left and center-left plots). Phase shift images of the sample objects were obtained with a digital holographic microscope. The equivalent diameter, the maximum optical height, the mean optical height and the optical volume of the sample objects were extracted from the phase shift images and are shown in FIG. **8**. Dissolving the sample objects using an enzymatic solution containing trypsin causes cell death and destruction of squamous cells, leading to a substantial decrease in the mean optical height and optical volume of the sample objects and thereby the conclusion of trypsin as unsuitable dissociation mean for squamous cells (center-left plot: no dissociation, center-right plot: treatment with trypsin for 3 min, right plot: treatment with trypsin for 5 min).

[0142] The embodiments of the present invention disclosed herein only constitute specific examples for illustration purposes. The present invention can be implemented in various ways and with many modifications without altering the underlying basic properties. Therefore, the present invention is only defined by the claims as stated below.

LIST OF REFERENCE SIGNS

[0143] **100**—microfluidic device [0144] **102**—sample [0145] **102A**—single cell [0146] **102B**—cell aggregate [0147] **102C**—piece of continuous tissue material [0148] **104**—substrate [0149] **104A**—inlet [0150] **104B**—outlet [0151] **104C**—inlet for dissolving/enzymatic solution [0152] **106**—insertion volume [0153] **108**—sorting unit [0154] **108A**—first outlet of sorting unit **108** [0155] **108B**—second outlet of sorting unit **108** [0156] **108C**—filter [0157] **110**—measurement volume [0158] **112**—optical detection window [0159] **114**—dissociation unit [0160] **114A-D**—dissociation volumes [0161] **116**—reservoir for dissolving/enzymatic solution [0162] **116A**—static obstacles (blades) [0163] **116B**—constricted portions [0164] **116C**—magnetic stirrer [0165] **116D**—acoustic/ultrasound emitter [0166] **118A-D**—pumps/valves [0167] **200A**, **200B**—microfluidic device [0168] **202**—sorter [0169] **202A**—first outlet of sorter **202** [0170] **202B**—second outlet of

sorter **202** [0171] **204, 204A, 204B**—filter [0172] **300**—microfluidic device [0173] **302**—sample collection unit [0174] **304**—control unit [0175] **306**—preparatory unit [0176] **308**—post-processing unit [0177] **600**—system for analyzing a sample comprising sample objects containing biological cells [0178] **602**—microfluidic device [0179] **604**—mount [0180] **606**—microfluidics unit [0181] **608**—measurement device (digital holographic microscope) [0182] **608A**—objective [0183] **608B**—holographic imaging system [0184] **608C**—imaging lens [0185] **608D**—camera [0186] **610**—controller [0187] **612**—acoustic/ultrasound source [0188] **700**—method for analyzing a sample comprising sample objects containing biological cells [0189] **702**—step of determining sample type [0190] **704**—step of generating carrier fluid flow [0191] **706**—step of sorting sample objects by size, weight and/or stiffness [0192] **708**—step of performing measurement on a batch of sorted sample objects [0193] **710**—step of dissociating sample objects in a residual sample of sorted sample objects

Claims

1-27. (canceled)

28. A microfluidic device for analyzing a sample comprising one or more sample objects containing biological cells, wherein the one or more sample objects comprise one or more of one or more single cells, one or more cell aggregates and one or more pieces of continuous tissue material, the microfluidic device comprising: an insertion volume configured to receive the sample; a sorting unit having an inlet that is in fluid communication with the insertion volume, a first outlet and a second outlet, wherein the sorting unit is configured to sort the sample objects by one or more of size, weight and stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, from the inlet towards the first outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, from the inlet towards the second outlet; a measurement volume that is in fluid communication with the first outlet of the sorting unit; and a dissociation unit comprising means for dissociating sample objects into one or both of single cells and cell aggregates at least in part, wherein an inlet of the dissociation unit is in fluid communication with the second outlet of the sorting unit and an outlet of the dissociation unit is in fluid communication with an inlet of the measurement volume.

29. The microfluidic device of claim 28, wherein the dissociation unit comprises a dissociation volume and the means for dissociating sample objects comprise one or more of: a reservoir configured to supply a dissolving solution to the dissociation volume, the dissolving solution comprising one or both of one or more chemical substances and one or more biochemical substances for chemically dissolving sample objects; an acoustic emitter configured to apply acoustic waves to sample objects in the dissociation volume; one or more static obstacles arranged in the dissociation volume configured to mechanically dissociate sample objects colliding with the one or more obstacles; one or both of one or more curved portions and one or more constricted portions of the dissociation volume configured to dissociate sample objects by generating a shear force on sample objects flowing through the dissociation volume; a stirrer configured to move a stir bar in the dissociation volume for dissociating sample objects; and one or more moveable elements configured to one or both of cut and grind sample objects in the dissociation volume.

30. The microfluidic device of claim 29, wherein the dissociation unit comprises a plurality of dissociation volumes, each of which is associated with a different means for dissociating sample objects.

31. The microfluidic device of claim 28, wherein one or more of: the outlet of the dissociation unit is in fluid communication with the measurement volume via the sorting unit; and the outlet of the dissociation unit further comprises a sorter having a first outlet that is in fluid communication with the measurement volume and a second outlet that is in fluid communication with the inlet of the dissociation unit, wherein the sorter is configured to sort the sample objects by one or more of size,

weight and stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, towards the first outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, towards the second outlet; and the microfluidic device further comprises a filter arranged between the outlet of the dissociation unit and the inlet of the measurement volume, wherein the filter is configured to prevent sample objects having a size larger than a filtering threshold from entering the measurement volume.

32. The microfluidic device of claim 28, wherein the microfluidic device comprises a sample collection unit comprising one or more of: a connector configured to couple one or both of a syringe and a tube containing a sample comprising the one or more sample objects to the insertion volume; an opening configured to receive one or both of an aspiration needle and a biopsy needle, wherein the opening is in fluid communication with the insertion volume; and a transfer pump configured to transfer a sample from a collection tube to the insertion volume.

33. The microfluidic device of claim 28, further comprising a flow pump configured to generate a flow through the insertion volume to transport the sample objects from the insertion volume to the sorting unit.

34. The microfluidic device of claim 28, wherein the measurement volume comprises a first measurement channel and a second measurement channel connected in parallel to the first measurement channel, the second measurement channel having a larger cross-sectional area than the first measurement channel.

35. The microfluidic device of claim 28, further comprising a measurement fluid inlet arranged between the outlet of the dissociation unit and the inlet of the measurement volume for supplying a measurement fluid.

36. The microfluidic device of claim 28, wherein the measurement volume comprises an optical detection window for taking microscopic images of sample objects in the measurement volume.

37. A method of analyzing a sample comprising one or more sample objects containing biological cells using a microfluidic device, wherein the one or more sample objects comprise one or more of one or more single cells, one or more cell aggregates and one or more pieces of continuous tissue material and the microfluidic device comprises: an insertion volume configured to receive the sample; a sorting unit having an inlet that is in fluid communication with the insertion volume, a first outlet and a second outlet, wherein the sorting unit is configured to sort the sample objects by one or more of size, weight and stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, from the inlet towards the first outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, from the inlet towards the second outlet; a measurement volume that is in fluid communication with the first outlet of the sorting unit; and a dissociation unit comprising means for dissociating sample objects into one or both of single cells and cell aggregates at least in part, wherein an inlet of the dissociation unit is in fluid communication with the second outlet of the sorting unit and an outlet of the dissociation unit is in fluid communication with an inlet of the measurement volume, the method comprising: generating a flow of a carrier fluid through the insertion volume to transport sample objects from the insertion volume to the sorting unit; sorting the sample objects with the sorting unit into a first batch of sample objects and a first residual sample, wherein the first residual sample comprises sample objects having one or more of a larger size, a larger weight and a higher stiffness than the sample objects in the first batch; performing a first measurement on the sample objects of the first batch in the measurement volume; dissociating sample objects in the first residual sample with the dissociation unit into one or both of single cells and cell aggregates at least in part to obtain a second batch of sample objects; and performing a second measurement on the dissociated sample objects in the second batch in the measurement volume.

38. The method of claim 37, wherein the second batch of sample objects is obtained by sorting the dissociated sample objects into the second batch of sample objects and a second residual sample, wherein the second residual sample comprises sample objects having one or both of a larger size, a

larger weight and a higher stiffness than the sample objects in the second batch.

39. The method of claim 38, further comprises obtaining one or more additional batches of sample objects by: dissociating sample objects in a residual sample separated from a previous batch into one or both of single cells and cell aggregates at least in part using the dissociation unit; and sorting the dissociated sample objects into the respective additional batch of sample objects and a new residual sample, wherein the new residual sample comprises sample objects having one or more of a larger size, a larger weight and a higher stiffness than the sample objects in the respective additional batch.

40. The method of claim 37, further comprising: determining a sample type of the sample, wherein the sample type characterizes one or more of a procedure used for obtaining the sample from a patient, a body part that the sample was obtained from, a size of the sample objects in the sample and a weight of the sample objects in the sample; determining a number of dissociation steps that are to be performed based on the sample type; and performing the dissociation steps and sorting the dissociated sample objects by one or more of size, weight and stiffness to obtain a plurality of batches of sample objects.

41. The method of claim 40, wherein the sample comprises one or more of a biopsy sample, a resection sample, an aspiration sample, a body fluid sample and a smear sample and wherein no dissociation step or a single dissociation step is performed one or both of if the sample type indicates that the sample is a body fluid sample and if the sample type indicates that the sample is a smear sample and multiple dissociation steps are performed one or more of if the sample type indicates that the sample is a biopsy sample, if the sample type indicates that the sample is a resection sample and if the sample type indicates that the sample is an aspiration sample.

42. The method of claim 40, wherein the dissociation unit of the microfluidic device comprises a plurality of means for dissociating the sample objects and the method further comprises selecting a means for dissociating the sample objects from the plurality of means for dissociating the sample objects based on the sample type.

43. The method of claim 37, wherein one or both of performing the first measurement and performing the second measurement comprises or corresponds to taking an image of the sample objects in the measurement volume using a microscope.

44. A system for analyzing a sample comprising one or more sample objects containing biological cells, the one or more sample objects comprising one or more of one or more single cells, one or more cell aggregates and one or more pieces of continuous tissue material, wherein the system is configured for use with a microfluidic device comprising: an insertion volume configured to receive the sample; a sorting unit having an inlet that is in fluid communication with the insertion volume, a first outlet and a second outlet, wherein the sorting unit is configured to sort the sample objects by one or more of size, weight and stiffness by directing sample objects having a smaller size, a smaller weight and a lower stiffness, respectively, from the inlet towards the first outlet and sample objects having a larger size, a larger weight and a higher stiffness, respectively, from the inlet towards the second outlet; a measurement volume that is in fluid communication with the first outlet of the sorting unit; and a dissociation unit comprising means for dissociating sample objects into one or both of single cells and cell aggregates at least in part, wherein an inlet of the dissociation unit is in fluid communication with the second outlet of the sorting unit and an outlet of the dissociation unit is in fluid communication with an inlet of the measurement volume, wherein the system comprises: a microfluidics unit configured to generate a flow of a carrier fluid through the insertion volume of the microfluidic device to transport sample objects from the insertion volume to the sorting unit of the microfluidic device; a measurement device configured to perform measurements on one or both of single cells and cell aggregates in the measurement volume of the microfluidic device; and a controller configured to control one or more of the microfluidics unit, the measurement device, the sorting unit of the microfluidic device and the dissociation unit of the microfluidic device to execute a method for analyzing the sample, said

method comprising: generating a flow of a carrier fluid through the insertion volume to transport sample objects from the insertion volume to the sorting unit; sorting the sample objects with the sorting unit into a first batch of sample objects and a first residual sample, wherein the first residual sample comprises sample objects having one or more of a larger size, a larger weight and a higher stiffness than the sample objects in the first batch; performing a first measurement on the sample objects of the first batch in the measurement volume; dissociating sample objects in the first residual sample with the dissociation unit into one or both of single cells and cell aggregates at least in part to obtain a second batch of sample objects; and performing a second measurement on the dissociated sample objects in the second batch in the measurement volume.

45. The system of claim 44, wherein the measurement device is or comprises a phase-contrast microscope configured to take phase shift images of one or both of single cells and cell aggregates in the measurement volume of the microfluidic device.

46. The system of claim 44, further comprising an acoustic source that is configured to apply acoustic waves to sample objects in the dissociation unit of the microfluidic device, wherein the controller is configured to control the acoustic source to dissociate sample objects in the dissociation unit by applying acoustic waves to the sample objects.

47. The system of claim 44, wherein the microfluidics unit is configured to provide a dissolving solution to the dissociation unit of the microfluidic device, the dissolving solution comprising one or both of one or more chemical substances and one or more biochemical substances for chemically dissolving sample objects, wherein the controller is configured to control the microfluidics unit to dissociate sample objects in the dissociation unit by supplying the dissolving solution to the dissociation unit.
